

LEARNING TO STEER, STEERING TO SEE: UNVEILING THE GEOMETRY OF RLVR IN LARGE LANGUAGE MODELS VIA TRAINABLE VECTORS

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ABSTRACT

Reinforcement learning (RL) has emerged as a key paradigm for enhancing the reasoning capabilities of large language models. However, the high dimensionality of parameter updates makes RL training dynamics difficult to analyze, obscuring the mechanisms underlying these gains. In this work, we use vector steering as an analytical tool to identify a low-dimensional effective manifold in activation space that is closely associated with RL-induced performance gains. We further uncover two key geometric properties of this manifold. **(1) Effective Manifold Capacity:** The capacity required to reproduce RL-induced gains can be very small, but it is not infinitely compressible. When the capacity is compressed to an extremely low level, the intervention dimensionality and the ability to express context-dependent corrections become key constraints. This requirement varies with injection depth and grows with the teacher-student distribution gap. **(2) Control Manifold Separation:** Effective control directions lie predominantly in the low-variance complement of the activation principal subspace. Within the same task and base model, the learned geometry remains largely consistent across training configurations; across tasks, geometric alignment correlates with capability transfer. Experiments on 5 LLMs and 6 tasks with verifiable rewards support these properties. Building on these findings, we propose **Alpha-Stabler**, a plug-and-play training framework comprising two modules: a **Predictor** that monitors principal-subspace intrusion to provide early warnings of training collapse, and a **Controller** that attenuates the intrusive component of RL-induced activation changes while preserving their off-principal components. Experiments show that Alpha-Stabler can stabilize training for 2,000 steps and consistently enhance the gains brought by reinforcement learning. Overall, this work advances the understanding of RL from an activation-manifold perspective, offering practical insights for more robust post-training.

“Mille viae ducunt homines per saecula Romam.”
— Alain de Lille

1 INTRODUCTION

Reinforcement learning (RL) has become central to post-training for large language models (LLMs), yielding substantial gains in reasoning performance (OpenAI, 2025; Xiao et al., 2026; Xu et al., 2026). Yet how RL produces these improvements internally remains incompletely understood. High-dimensional parameter updates, optimization noise, and sensitivity to training configurations make it difficult to distinguish behaviorally meaningful changes from incidental variation. Prior work has examined sampling efficiency (Yue et al., 2025), entropy dynamics (Cui et al., 2025b), scaling behavior (Tan et al., 2026), and update geometry (Cai et al., 2025; 2026; Zhu et al., 2025). A complementary question remains unanswered: what structure do RL-induced behavioral changes exhibit in activation space, and how does this structure relate to capability transfer and training stability?

We view RL post-training as an intervention problem in activation space. Concretely, we freeze the base model and learn input-invariant shared vectors at selected layers to distill its RL-tuned counterpart, thereby compressing RL’s high-dimensional parameter updates into low-dimensional activation changes. This directly tests whether RL gains can be recovered by low-dimensional vectors and when this low-dimensional description breaks down. We thereby identify a low-dimensional effective manifold in activation space tied to RL-induced gains, and characterize two geometric properties of this manifold: **Effective Manifold Capacity** and **Control Manifold Separation**.

Our first finding concerns the effective capacity of this manifold. With an RL-trained model as the teacher, a single input-invariant vector per controlled layer recovers over 85% of the gain achieved by full fine-tuning on most evaluated tasks, as shown in Fig 1 (a) left. However, this compressibility has clear limits. The first arises from the **teacher-student distribution gap**: when the two distributions are close, a single-vector intervention suffices, whereas a widening gap requires more flexible parameterizations. The second arises from **insertion depth**: fixed-vector interventions work well at low-to-middle layers but degrade near the output, as shown in Fig 1(a)

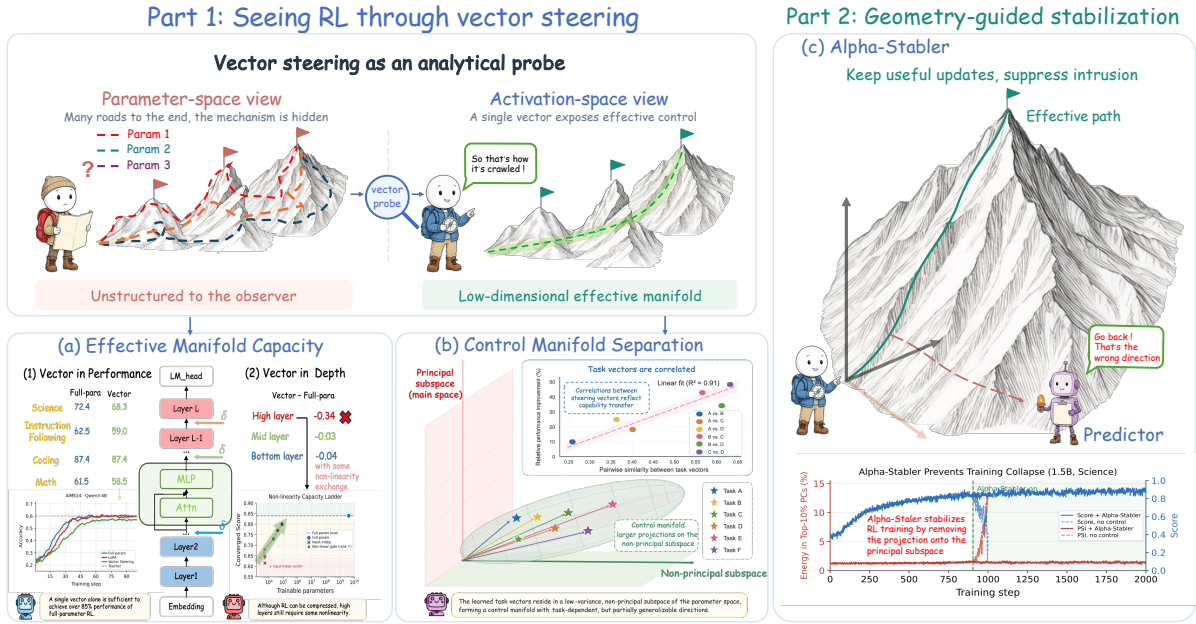


Figure 1: The single steering vector reveals that RLVR-induced changes form a low-dimensional, depth-sensitive control geometry, characterized by (a) effective manifold capacity and (b) control-manifold separation. Building on these findings, (c) Alpha-Stabler stabilizes training via principal-subspace intrusion monitoring and backward gradient projection.

right. Through nonlinear analysis, we further find that this degradation stems not from weak optimization signals at deeper layers, but from an expressiveness bottleneck of high-layer interventions. We summarize these findings as **Property 1 (Effective Manifold Capacity)**: the capacity required to reproduce RL-induced gains can be very small, but it is not infinitely compressible; this capacity depends on how well the intervention form matches the required correction.

We next examine the geometry of the effective manifold. By extending the single-vector probe to multiple mutually orthogonal vectors per layer, we find that although the standalone performance of later directions declines with extraction order, they can still independently support the target behavior, indicating that the leading direction is dominant but not uniquely effective. While RL’s training parameters differ substantially across configurations in parameter space, they consistently converge, after compression, to a set of reproducible task-related directions. Across tasks, the alignment between direction vectors strongly reflects transfer gains, and these directions consistently concentrate in the low-variance complement of the activation principal subspace, as shown in Fig 1 (b). We characterize this organization as **Property 2 (Control Manifold Separation)**: RL-induced control converges to a set of reproducible task-related directions that are largely separated from the variance-dominant representation subspace.

These two properties jointly indicate that RL-induced gains are not high-dimensional disorder, but can be compressed into a set of low-dimensional directions; successful RL training confines this update to the low-variance complement of the principal subspace, whereas principal-subspace intrusion accompanies performance degradation and collapse. To this end, we propose **Alpha-Stabler**, a lightweight, plug-and-play framework for stabilizing RL training. It introduces no additional trainable parameters and requires no modification to the optimizer, reward function, or model architecture. Alpha-Stabler consists of a **Predictor** that monitors principal-subspace intrusion (PSI) against a fixed principal subspace from the frozen base model and warns before collapse, and a **Controller** that applies a low-rank projection to activation gradients at flagged layers during backpropagation, removing the intrusive component while preserving the effective complement update. Experiments show that, with less than 2% additional computation and zero additional trainable parameters, Alpha-Stabler achieves more stable reward growth over thousands of training steps than gradient clipping and KL regularization.

In summary, this work uncovers two geometric laws governing RL’s effect on LLMs: Effective Manifold Capacity and Control Manifold Separation. These laws reveal that the seemingly complex dynamics of RL post-training are governed by a surprisingly simple structure: a low-dimensional manifold confined to the low-variance complement of the activation principal subspace. This challenges the black-box view of RL and opens new avenues for bridging empirical observations and theoretical understanding of capability emergence. Building on these laws, we introduce **Alpha-Stabler**, a plug-and-play framework that operates with negligible overhead and provides a practical tool for stabilizing large-scale RL training.

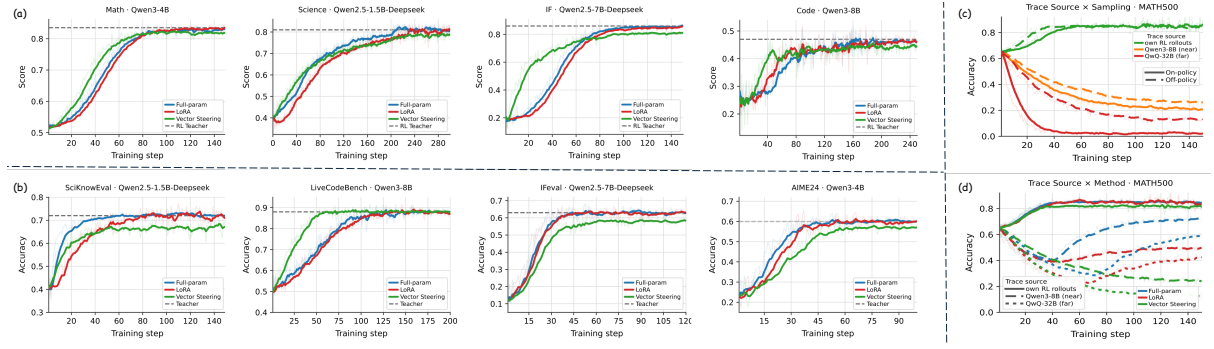


Figure 2: (a) On-policy distillation scores on the training set. (b) Off-policy distillation scores on the evaluation set. (c) Scores of vector steering under varying teacher types. (d) Scores of full-parameter training, LoRA, and vector steering under off-policy distillation at three distribution gaps.

2 EFFECTIVE MANIFOLD CAPACITY

2.1 EXPERIMENTAL SETTING

Models and Datasets. We evaluate DeepSeek-R1-Distill-Qwen-1.5B/7B (Guo et al., 2025) and Qwen3-4B/8B/14B (Yang et al., 2025) on four task families: mathematical reasoning, scientific reasoning, code generation, and instruction following. For each task, we train a teacher with GRPO or DAPO and perform on- or off-policy distillation (Li et al., 2026b), comparing three update parameterizations: full-parameter training, LoRA (Hu et al., 2021), and vector steering. Details on datasets, sampling, optimization, and prompts are provided in Appendix B.2.

Vector Steering. Consider a frozen base model with L decoder layers and hidden dimension d . For a set of controlled layers $\mathcal{S} \subseteq \{1, \dots, L\}$, we add a trainable vector to each layer’s output:

$$\mathbf{h}'_{\ell} = \mathbf{h}_{\ell} + \delta_{\ell}, \quad \ell \in \mathcal{S}, \quad (1)$$

where $\mathbf{h}_{\ell} \in \mathbb{R}^d$ is the hidden state at a given token position, and $\delta_{\ell} \in \mathbb{R}^d$ is a layer-specific vector shared across all inputs and token positions. Only $\{\delta_{\ell}\}_{\ell \in \mathcal{S}}$ are optimized under the corresponding distillation objective. We then examine the compressibility of RL updates in Section 2.2 and further analyze its properties in Section 2.3.

2.2 COMPRESSIBILITY OF RL LEARNING SIGNALS

Compressibility across Distillation Regimes. We examine how much of the capability gain obtained by RL can be retained after distillation under restricted update parameterizations. As shown in Figure 2 (a), under on-policy distillation, LoRA and full-parameter training perform comparably, while a single steering vector per controlled layer still recovers over 85% of the RL gain and nearly matches full-parameter training on Qwen3-8B code generation. To test whether this compressibility depends on student-generated trajectories (Yang et al., 2026), we also evaluate off-policy distillation with teacher-generated trajectories under the same update parameterizations. As shown in Figure 2 (b), steering retains a comparably high recovery rate on evaluation datasets. Together, these results indicate that RL-induced gains can be highly compressed, and that fixed layer-wise activation offsets alone suffice to retain most of them.

Effect of Teacher Type and Update Capacity. These recovery rates, however, have limits. We fix the student and vary the teacher under both distillation regimes to examine the sensitivity of vector steering to the teacher–student distribution gap. As shown in Figure 2 (c), vector steering effectively transfers the capabilities of RL-trained teachers, whereas tested non-RL teachers degrade substantially, with some settings collapsing as the gap widens. Teacher–student distribution matching is therefore a key factor governing the effectiveness of fixed activation offsets.

A natural explanation is that the required corrections become more input-dependent as the gap widens, exceeding what an input-invariant fixed offset can express. If so, parameterizations with different expressive capacities should exhibit a consistent ordering. As shown in Figure 2 (d), under off-policy distillation, the three parameterizations perform comparably when the gap is small, whereas a larger gap yields a clear ordering: full-parameter training performs best, followed by LoRA, and vector steering worst. This ordering matches their expressive capacities—full-parameter training allows model-wide adjustments, LoRA retains input-dependent corrections under low-rank constraints, and vector steering only applies fixed offsets. A larger gap therefore raises the input dependence required of the correction, and fixed additive corrections are the first to reach their capacity limit (Cai et al., 2026).

Implications for RL Compressibility. Together, although vector steering cannot cover arbitrary teacher-induced corrections, it serves as a targeted probe for analyzing the mechanisms underlying RL-specific capability gains.

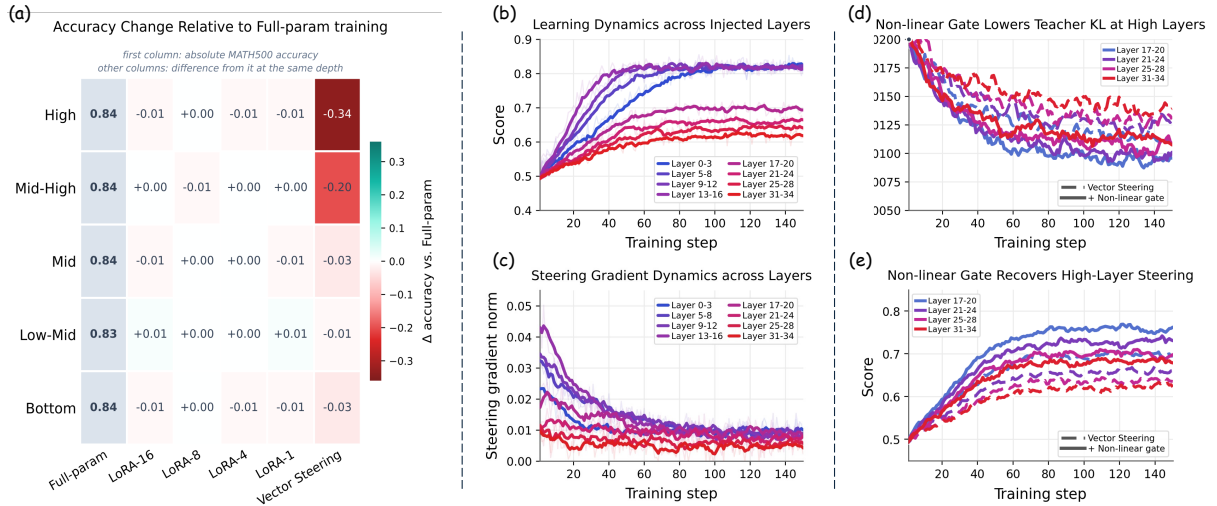


Figure 3: Layer-wise performance. (a) Converged performance of steering vectors across injection layers. (b) Task performance during training for steering vectors. (c) Steering gradient norms at different layers. (d) KL divergence with the teacher under fixed versus gated steering. (e) Converged performance of fixed versus gated injection.

Building on this, we next examine how RL-induced gains vary with injection depth and how effective the low-dimensional representation is across layers.

2.3 LAYER-WISE COMPRESSIBILITY OF RL UPDATES

Injection Depth and Optimization Dynamics. We compare full-parameter updating, LoRA, and fixed-vector steering at representative network depths, restricting each intervention to the selected layer and freezing the remaining backbone. As shown in Figure 3 (a), selected-layer full-parameter updating and LoRA remain effective across depths, whereas fixed-vector steering performs well at lower and intermediate layers but deteriorates near the output. This contrast indicates that low-capacity interventions depend on both the update parameterization and the injection depth.

We next examine gradient norms and learning curves across injection sites. Figure 3 (b) shows that shallow-layer steering gradually approaches intermediate-layer performance during training, whereas high-layer steering plateaus early, and extended training does not noticeably close the gap. To test whether this stagnation stems from weaker gradient signals, we inspect gradient norms. Figure 3 (c) further shows that shallow and high-layer injections receive gradients of comparable magnitude yet yield substantially different final performance, indicating that gradient magnitude alone does not explain high-layer degradation. A possible explanation is that fewer downstream nonlinear transformations at high layers limit the ability of a fixed offset to produce context-dependent effects, whereas LoRA and selected-layer full-parameter updates can directly produce input-dependent activation corrections.

Input-Dependent Corrections and Capacity Controls. To test whether context-dependent modulation alleviates high-layer degradation, we replace the fixed offset with a low-rank nonlinear correction:

$$\mathbf{h}'_\ell = \mathbf{h}_\ell + \delta_\ell(\mathbf{h}_\ell), \quad \delta_\ell(\mathbf{h}_\ell) = \mathbf{B}_\ell \sigma(\mathbf{A}_\ell \mathbf{h}_\ell), \quad (2)$$

where $\mathbf{A}_\ell \in \mathbb{R}^{r \times d}$, $\mathbf{B}_\ell \in \mathbb{R}^{d \times r}$, $r \ll d$, and σ is an element-wise gating nonlinearity. Here $\mathbf{B}_\ell$ defines shared correction directions and $\sigma(\mathbf{A}_\ell \mathbf{h}_\ell)$ assigns token-dependent coefficients. We zero-initialize $\mathbf{B}_\ell$ so that the intervention initially leaves the model unchanged. This adds $O(rd)$ trainable parameters while keeping the correction within a shared low-dimensional subspace. Figure 3 (d)–(e) shows that gated steering maintains lower teacher KL divergence and achieves higher converged performance at all four tested high-layer injection sites.

To separate input dependence from parameter count, we vary the gating rank and include a parameter-matched input-independent control. Figure 4 (a) shows that rank-1 gating substantially outperforms fixed-vector steering, whereas the static control provides no improvement over the baseline. Increasing the rank yields further gains, approaching full-parameter updating of the selected layer at $r = 16$. Recovery therefore stems from token-dependent modulation, not parameter count alone.

Directional Structure and Effective Manifold Capacity. We further examine how correction demands and learned modulation vary across depth. At each layer, we collect the output-gradient directions at all token positions and compute the mean pairwise cosine similarity over token pairs, $\text{Sim}_\ell^{\text{grad}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \cos(\mathbf{g}_{\ell,i}, \mathbf{g}_{\ell,j})$, where $\mathbf{g}_{\ell,i}$ is the output-gradient direction at layer ℓ and token position i , and $\mathcal{P}$ is the set of token pairs. As a reference, we compute the same average pairwise cosine similarity for hidden states. Figure 4 (b) shows that the mean similarity

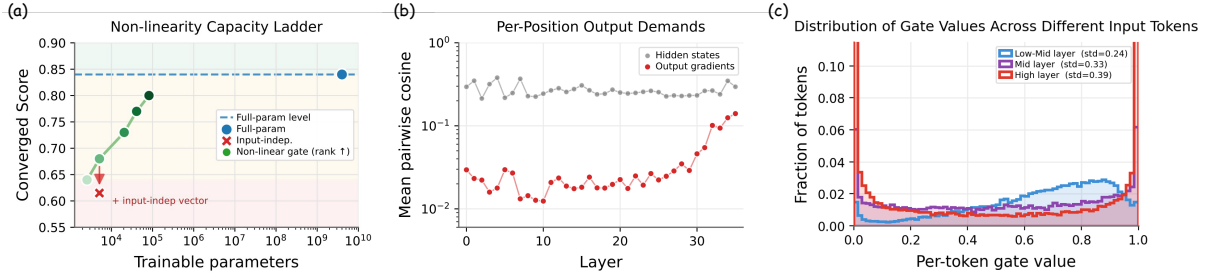


Figure 4: Diagnosing and mitigating high-layer degradation. **(a)** Performance of gated injection at high layers as a function of the gating rank r . **(b)** Mean pairwise cosine similarity across layers between per-token output-update directions and per-position gradient-update directions. **(c)** Distribution of gating activations at different layers.

of gradient directions is approximately 0.019 at lower and intermediate layers and rises to approximately 0.14 near the output. This trend is inconsistent with the degradation of fixed-vector steering at high layers: the more aligned the directions, the worse the performance, indicating that average directional alignment alone does not explain steering effectiveness. This result suggests that what matters is not whether directions align, but how the correction varies across tokens. To test this, Figure 4 (c) examines the distribution of the post-activation gating coefficient $\sigma(\mathbf{A}_\ell \mathbf{h}_\ell)$ across tokens for rank-1 gating. A global rescaling would leave this distribution depth-invariant, whereas token-dependent modulation would not. At lower and intermediate layers, the coefficients are broadly distributed over $[0, 1]$, consistent with graded adjustment of correction strength. At higher layers, they concentrate near 0 and 1 while remaining heterogeneous across tokens, indicating increasingly selective application of the shared direction rather than convergence to an input-independent correction.

Summary. We summarize these experimental findings as **Property 1 (Effective Manifold Capacity)**: reproducing RL-induced gains requires only a very low-dimensional intervention, but its effective capacity depends on how well the intervention form matches the required correction. In Appendix D.4, we provide a conditional theoretical account of fixed-vector recovery as a balance between shared teacher-induced shifts and their context-dependent fluctuations.

3 CONTROL MANIFOLD SEPARATION

Having established strong recovery with one steering vector per controlled layer, we now examine the diversity, reproducibility, and principal-subspace localization of effective steering directions.

3.1 PROBING RESIDUAL EFFECTIVE DIRECTIONS WITH MULTI-VECTOR STEERING

Sequential orthogonal steering. To test whether the leading direction is uniquely effective, we sequentially learn K vectors $\mathcal{V}_\ell = \{\mathbf{v}_\ell^{(k)}\}_{k=0}^{K-1}$ at each controlled layer $\ell \in \mathcal{S}$, enforcing pairwise orthogonality against earlier vectors, $\langle \mathbf{v}_\ell^{(k)}, \mathbf{v}_\ell^{(j)} \rangle = 0$ for $j < k$. The first stage is trained to convergence and frozen; each subsequent stage initializes new vectors in the remaining orthogonal complement, optimizes them to convergence, and reprojects them after every optimizer step:

$$\mathbf{v}_\ell^{(k)} \leftarrow \mathbf{v}_\ell^{(k)} - \sum_{j=0}^{k-1} \frac{\langle \mathbf{v}_\ell^{(k)}, \mathbf{v}_\ell^{(j)} \rangle}{\|\mathbf{v}_\ell^{(j)}\|_2^2} \mathbf{v}_\ell^{(j)}. \quad (3)$$

Only the current stage’s vectors are injected, jointly defining a multi-layer intervention, while earlier vectors remain frozen and serve solely to define the orthogonality constraints. At each stage, only the vectors learned at that stage are injected and evaluated in isolation.

Residual effectiveness and model scale. As shown in Figure 5 (a), the first direction achieves 67.5% standalone accuracy on SciKnowEval with Qwen2.5-1.5B-DeepSeek, while the eighth direction, $\mathbf{v}^{(7)}$, achieves 62.0%, only 5.5 percentage points lower despite being orthogonal to the preceding seven directions at every controlled layer. This modest reduction indicates that excluding previously learned directions does not exhaust the available steering solutions: the remaining orthogonal space still admits interventions with substantial standalone effectiveness. Training results across all four task domains are shown in Appendix Figure 11.

We further examine how this phenomenon varies with model scale. As shown in Figure 5 (b), the leading direction consistently performs best, while standalone accuracy generally declines with extraction order; the decline is sharper in smaller models, whereas larger models retain higher-order directions whose performance remains closer

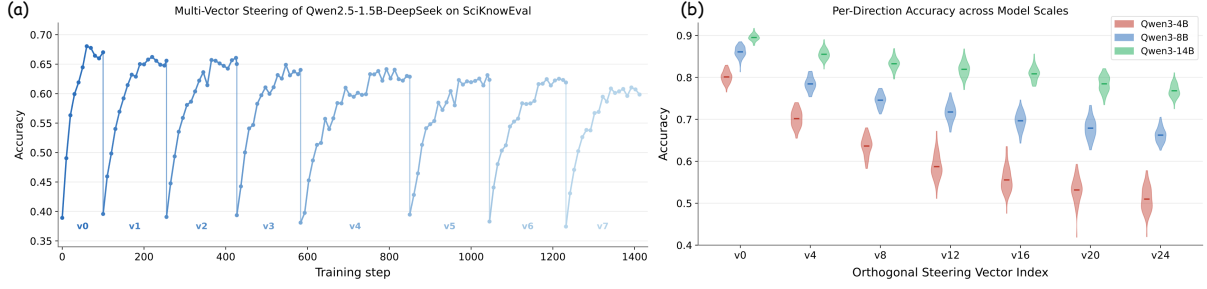


Figure 5: Multi-vector steering with sequential orthogonalization. **(a)** Training trajectories of sequentially learned orthogonal steering vectors on SciKnowEval with Qwen2.5-1.5B-DeepSeek. **(b)** Standalone accuracy of the extracted directions across model scales. Each direction is trained and evaluated independently, with preceding directions used only to define the orthogonality constraint.

to that of the leading direction. This indicates that model scale affects not only the performance of the leading direction, but also the extent to which higher-order directions are retained relative to it.

Summary. These results indicate that the RL-induced gains probed by steering are not concentrated in a single dominant direction, but are distributed across a set of ordered, mutually orthogonal directions: multiple orthogonal interventions can independently support the target behavior, although their standalone effectiveness declines systematically with extraction order. We next examine whether this ordered geometry is recovered consistently across teacher parameterizations and distillation regimes.

3.2 TASK-DEPENDENT GEOMETRY AND PRINCIPAL-SUBSPACE SEPARATION

Consistency across training configurations. We next ask whether the extracted steering directions are reproducible across different training configurations. To this end, we compare directions learned under two axes of variation: teacher parameterizations, where we vary the teacher checkpoint used to generate supervision, e.g., teachers obtained by full-parameter fine-tuning or LoRA with different ranks, and distillation distributions, where we compare on-policy and off-policy distillation, which use student-generated trajectories and teacher-generated responses, respectively, thereby exposing steering to different sequence and activation distributions. For each axis, we extract multi-vector directions following the same procedure and measure similarity between two sets of directions using the layer-averaged cosine similarity between their ordered vectors, $\text{sim}(\{\mathbf{v}_a^{(i)}\}, \{\mathbf{v}_b^{(j)}\}) = \frac{1}{|\mathcal{S}|} \sum_{\ell \in \mathcal{S}} \cos(\mathbf{v}_{a,\ell}^{(i)}, \mathbf{v}_{b,\ell}^{(j)})$, where $\mathcal{S}$ denotes the controlled layer set. As shown in Figure 6 (a) and (b), directions extracted under different teacher parameterizations and under on-policy versus off-policy distillation remain closely aligned, with similarity concentrated along the diagonal of the ordered vector pairs, i.e., alignment between directions sharing the same extraction index is stronger than between different indices. This indicates that the extracted directions are not artifacts of a particular training configuration or distillation regime, but reflect a reproducible geometric structure shared across configurations: capability improvements correspond to an ordered set of effective directions that can be stably recovered.

Cross-topic alignment and capability transfer. Having established within-task consistency, we next ask whether differences between task-specific directions relate to capability transfer. We partition the Science training and test sets into physics, chemistry, materials, and biology. Keeping the base model fixed, we train one RL teacher on each topic’s training subset and distill it with the same procedure to extract its multi-vector directions, testing whether vector geometry tracks functional relationships across capabilities.

For source topic s and target topic t , we define directional alignment as $A(s, t) := C_{s,t}^{(0,0)}$, and normalize the transfer gain by the gain from training directly on the target topic:

$$\Delta_{\text{rel}} \text{Acc}(s \rightarrow t) = \frac{\text{Acc}(M_s, t) - \text{Acc}(M_{\text{base}}, t)}{\text{Acc}(M_t, t) - \text{Acc}(M_{\text{base}}, t)} \times 100\%, \quad (4)$$

where M_s , M_t , and M_{base} denote the source-trained, target-trained, and base models, respectively. As shown in Figure 6 (c), topic pairs with stronger directional alignment tend to exhibit larger relative transfer gains: a linear fit over the six ordered pairs of distinct topics yields an R^2 of 0.91. Thus, capability gains across topics are not isolated but share comparable effective directions: although parameter changes may differ, the induced activation-space directions remain geometrically comparable, and stronger alignment predicts more effective transfer.

Effective steering directions avoid the principal subspace. Having established that the directions are reproducible across training configurations, we next examine where they lie in activation space. Let $U_{\ell,r} \in \mathbb{R}^{d \times r}$ contain the leading orthonormal eigenvectors of the token-level activation covariance at layer ℓ , with r corresponding to

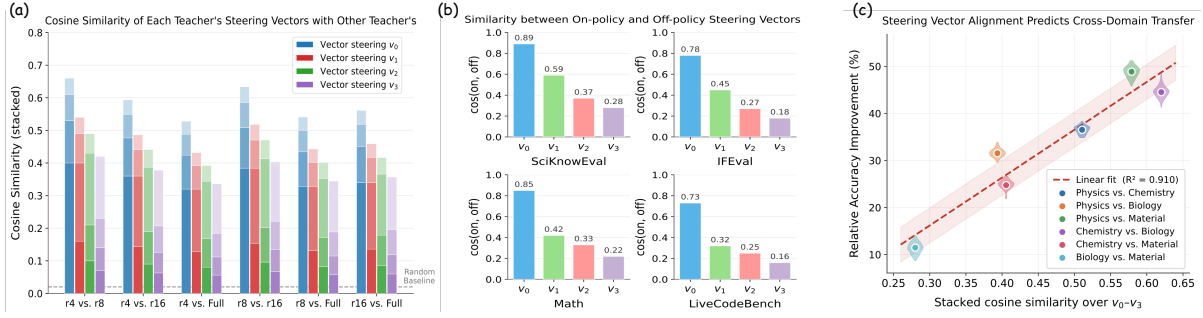


Figure 6: Geometric consistency and transfer of steering directions. **(a, b)** Layer-averaged cosine similarities between sequentially learned orthogonal vectors, across teacher parameterizations **(a)** and distillation regimes **(b)**. In **(a)**, a and b denote the first and second teacher parameterizations in each pair; in **(b)**, they denote on-policy and off-policy distillation. Each stacked bar corresponds to a reference vector $\mathbf{v}_a^{(i)}$; its segments, ordered from $\mathbf{v}^{(0)}$ to $\mathbf{v}^{(3)}$, indicate similarities with $\{\mathbf{v}_b^{(j)}\}_{j=0}^3$. **(c)** Cross-topic directional alignment versus relative accuracy improvement over the base model, with a fitted regression line.

10% of the hidden dimension, and define $P_\ell = U_{\ell,r}U_{\ell,r}^\top$. For a layer-wise direction $\mathbf{z}_\ell$, its principal-subspace energy fraction is

$$e_{\text{PC}}(\mathbf{z}_\ell) = \frac{\|P_\ell \mathbf{z}_\ell\|_2^2}{\|\mathbf{z}_\ell\|_2^2 + \epsilon}. \quad (5)$$

For a multi-layer direction, we average e_{PC} across the controlled layers. As shown in Figure 7 (a), across the four scientific topics, unconstrained steering places only about 1%–2% of its energy in the top-10% principal subspace, well below the 10% isotropic reference. This pattern persists from $\mathbf{v}^{(0)}$ through $\mathbf{v}^{(8)}$ and across all extraction orders, indicating that low principal-subspace energy is a shared geometric property of the extracted interventions.

Restricting steering to the principal subspace does not recover the gain. To test the functional importance of the low-variance complement, we remove each steering vector’s projection onto the bottom-70% activation-PC subspace after every optimizer step, restricting it to the span of the top-30% components: $\mathbf{v}_\ell^{(k)} \leftarrow P_{\ell,m} \mathbf{v}_\ell^{(k)}$. As shown in Figure 7(b), the top-10% energy fraction rises to approximately 15%, but accuracy does not improve meaningfully. The gain therefore does not come from alignment with the principal subspace.

Principal-subspace intrusion tracks training collapse. We next ask whether unconstrained RL training stays outside this subspace, and whether it departs from it when training becomes unstable. At training step t , we run the current policy and the frozen base model on the same token sequences, defining the activation shift $\Delta_{\ell,b,i}^{(t)} = \mathbf{h}_{\ell,b,i}^{(t)} - \mathbf{h}_{\ell,b,i}^{0,(t)}$ for sample b , position i , and layer ℓ . Since these shifts vary across examples and token positions, we stack all N valid token-level shifts row-wise into $X_\ell^{(t)} \in \mathbb{R}^{N \times d}$ to avoid cancellation before projection, and compute

$$\text{PSI}_\ell^{(t)} = \frac{\|X_\ell^{(t)} U_{\ell,r}\|_F^2}{\|X_\ell^{(t)}\|_F^2 + \epsilon}, \quad (6)$$

where $U_{\ell,r}$ is the fixed top-10% principal-subspace basis of the base model. As shown in Figure 7(c), collapse-prone runs exhibit a clear pattern: during stable training, PSI remains around 2%–3%, well below the 10% isotropic reference, while the task score improves or stays near its maximum; as training proceeds, PSI departs from this range and becomes more volatile before visible score degradation; as collapse progresses, PSI continues to rise while the score falls more sharply.

Together, these results support **Property 2 (Control Manifold Separation)**: effective steering directions predominantly occupy the low-variance complement of the activation principal subspace, their ordered geometry is reproducible across teacher parameterizations and distillation regimes, and cross-topic alignment tracks transfer gains. We provide a conditional theoretical account in Appendix D.4.2.

4 ALPHA-STABLER: GEOMETRY-GUIDED RL STABILIZATION

Experiments show that effective steering directions lie in the low-variance complement, while principal-subspace intrusion accompanies training collapse. We therefore propose Alpha-Stabler to monitor whether the RL-induced activation shift departs from this regime. It consists of a Predictor, which detects sustained increases in principal-subspace intrusion (PSI), and a Controller, which removes the principal-subspace component of activation gradients during backpropagation while preserving the low-variance complement.

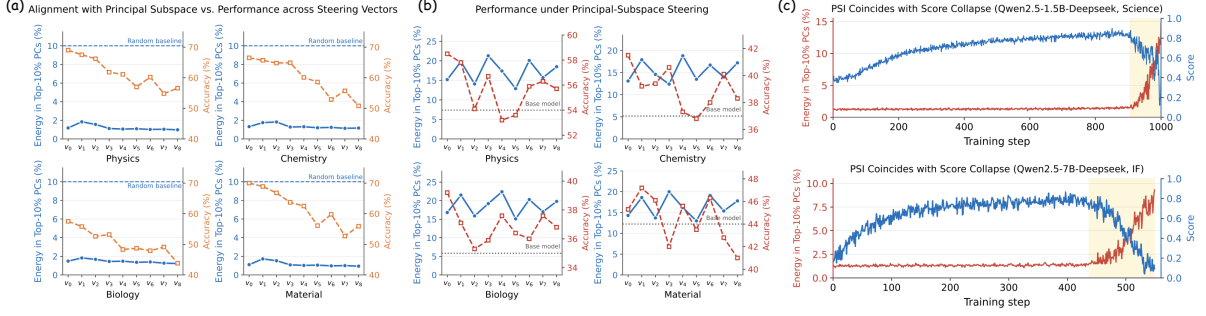


Figure 7: Steering geometry and principal-subspace intrusion. Energy fractions in (a) and (b) use the top-10% principal subspace. (a) Standalone accuracy and principal-subspace energy fractions of sequentially extracted directions across four scientific topics. (b) The same quantities when steering vectors are restricted to the top-30% activation-PC subspace. (c) Task score and PSI during RL training, comparing stable and collapse-prone runs.

4.1 ONLINE MONITORING AND CONTROL

At parameter-update step t , we evaluate the actor and the frozen initial base model on identical token sequences. We stack their valid token-level differences, $\Delta_{\ell,b,i}^{(t)} = \mathbf{h}_{\ell,b,i}^{(t)} - \mathbf{h}_{\ell,b,i}^{(0,t)}$, into $X_{\ell}^{(t)} \in \mathbb{R}^{N \times d}$. During the first $T_{\text{warm}} = 50$ updates, actor-generated rollouts are evaluated by the frozen base model to estimate the base model’s token-level activation covariance at each monitored layer. At the end of warm-up, we extract its leading orthonormal eigenvectors $U_{\ell,r}$, with $r = \lceil 0.10d \rceil$, and keep these bases fixed thereafter. Using the layer-wise PSI defined in Eqn. 6, we additionally aggregate across monitored layers:

$$\text{PSI}^{(t)} = \frac{\sum_{\ell \in \mathcal{S}} \|X_{\ell}^{(t)} U_{\ell,r}\|_F^2}{\sum_{\ell \in \mathcal{S}} \|X_{\ell}^{(t)}\|_F^2 + \epsilon}. \quad (7)$$

Layer-wise PSI determines where control is activated; the aggregate summarizes intrusion across monitored layers according to their shift energies. Both sum token-level energies before normalization, avoiding cancellation between differently oriented shifts. PSI is computed from detached activations, so it does not affect gradients, while the actor’s computation graph is retained for backpropagation.

Predictor: detecting sustained intrusion. During an initially stable warm-up, the actor trains without projection, and valid activation-shift matrices are retained every M updates. Once the principal bases are fixed, we compute PSI for these retained shifts and set fixed warning and lower release thresholds from layer-wise medians and scaled median absolute deviations. Subsequent valid observations update an exponential moving average initialized at the warm-up median. Control activates after p consecutive checks above the warning threshold and releases below the lower threshold. Algorithm 1 specifies the full procedure.

Controller: projecting activation gradients. The Controller modifies backward gradients without changing forward activations. Let $G_{\ell}^{(t)} \in \mathbb{R}^{N \times d}$ denote the incoming gradient at layer ℓ , including any downstream control effects. With the Predictor’s flag $a_{\ell}^{(t)} \in \{0, 1\}$ fixed during backpropagation and the optimizer step, we apply

$$\tilde{G}_{\ell}^{(t)} = G_{\ell}^{(t)} - a_{\ell}^{(t)} (G_{\ell}^{(t)} U_{\ell,r}) U_{\ell,r}^{\top}. \quad (8)$$

When active, this projection removes each token’s principal gradient component while preserving its orthogonal complement. This modification integrates directly into standard RL training: rollout generation, likelihood ratios, rewards, and the GRPO or DAPO loss remain unchanged, and the optimizer updates parameters using only the modified backward signal. Warm-up requires estimating reference statistics and temporarily storing calibration shifts; after calibration, reference activations are needed only at monitoring checks and can be reused from a compatible pass. Each active hook performs two low-rank matrix multiplications, and inference uses the trained actor alone, so the additional computation of Alpha-Stabler is minimal. Algorithm 1 summarizes the full training loop, including when flags are updated and when gradient projection is applied.

4.2 EXPERIMENTAL EVALUATION

Training stabilization. Figure 8 (a) and (b) compare training trajectories with and without Alpha-Stabler. In uncontrolled runs, a pronounced increase in PSI coincides with sharp score deterioration, indicating that performance loss accompanies a departure from the earlier activation geometry. Under control, PSI remains near its early-training range while task scores continue to improve, yielding sustained learning alongside limited principal-subspace intrusion rather than reduced PSI at the expense of task performance.

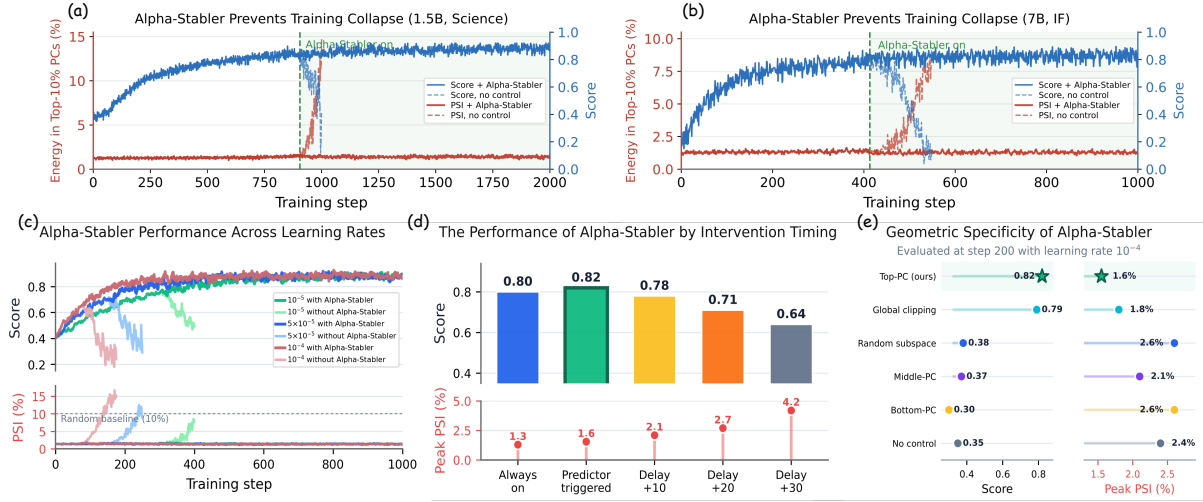


Figure 8: **(a, b)** Training dynamic with and without Alpha-Stabler. **(c)** Training dynamics across learning rates. **(d)** Task score and peak PSI under different intervention timings. **(e)** Performance under control of different subspaces.

Ablation studies. We then examine the ablation performance of Alpha-Stabler along three dimensions: learning rate, intervention timing, and controlled subspace. **First**, we find that stabilization holds across learning rates, as shown in Figure 8 (c): at any learning rate, uncontrolled training eventually collapses, whereas controlled training keeps peak PSI within a low single-digit percentage and reaches comparable final scores across settings, indicating robustness to the choice of learning rate. **Second**, triggered control is preferable to delayed control, as shown in Figure 8 (d). Triggered control matches always-on control in both score and peak PSI, indicating that continuous intervention is unnecessary, whereas delaying intervention progressively degrades the score and raises peak PSI. **Third**, Top-PC projection outperforms alternative subspaces, as shown in Figure 8 (e). Among the compared control subspaces, Top-PC projection achieves the highest score, while random-subspace, Middle-PC, and Bottom-PC control fail to match it and remain close to the uncontrolled baseline; Top-PC also compares favorably with global gradient clipping. This indicates that the improvement does not arise from an arbitrary form of gradient constraint, but from removing a specific direction, namely the principal subspace. These results support geometrically targeted control rather than indiscriminate gradient suppression. Appendix D.6.1 establishes the minimal-change property of gradient projection and derives a coupling-dependent local bound on principal-subspace activation leakage under projected updates. We further report wall-clock training time with and without Alpha-Stabler in Appendix 12; the results are nearly identical, with no measurable increase in the evaluated settings.

5 LIMITATIONS AND FUTURE WORK

Despite revealing two geometric laws governing RLVR’s effect on LLMs, our study has several limitations. First, the theoretical framework rests on assumptions that are empirically motivated but not yet derived from first principles. Second, our analysis and experiments are restricted to RLVR with verifiable rewards, such as mathematical reasoning and code generation, and we do not validate the proposed geometry on RLHF, multi-turn agentic decision-making, or open-ended generation; whether the same laws hold in these settings remains open. Future work can relax these assumptions via neuron attribution and causal tracing, extend the geometric framework to RLHF and agentic settings, and exploit the off-principal localization property as a broader design principle. For instance, one may add complement-space regularizers or use PSI as an adaptive signal for learning rate and intervention strength, enabling low-rank control without layer monitoring.

6 CONCLUSION

In this work, we uncover two geometric laws governing RLVR’s effect on LLMs: Effective Manifold Capacity, where reproducing RLVR-induced gains requires only a very low-dimensional intervention whose capacity depends on how well the intervention form matches the required correction; and Control Manifold Separation, where RLVR-induced control converges to reproducible task-related directions concentrated in the low-variance complement of the activation principal subspace. Building on these laws, Alpha-Stabler monitors principal-subspace intrusion as an early-warning signal and projects away the intrusive component of activation gradients during backpropagation, stabilizing training with almost no additional computational overhead. Overall, this work shifts the understanding of RLVR from black-box parameter optimization to an interpretable geometric process, opening new avenues for theoretically grounded and practically efficient LLM post-training.

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A RELATED WORK

Reinforcement Learning for Large Language Models. Prior to reasoning-capable models like OpenAI’s o1, RL was primarily used in RLHF for instruction-following and human preference alignment (Ouyang et al., 2022). Recently, Reinforcement Learning with Verifiable Rewards (RLVR) has proven effective for reasoning in verifiable domains such as mathematics and programming (Lambert et al., 2025). OpenAI’s o1 inspired a wave of open reasoning models, including DeepSeek-V4 (Xu et al., 2026), Kimi-K3 (Team et al., 2026), Qwen3.8 (Qwen Team, 2026), and GLM-5.3 (Z.ai, 2026). Beyond single-turn reasoning, RL is increasingly applied to agentic settings involving planning, tool invocation (Li et al., 2026a), and multi-turn environment interaction (Ding et al., 2026), with notable efforts on search-augmented reasoning (Jin et al., 2025), multi-turn credit assignment (Zhou et al., 2025), and asynchronous large-scale agentic RL (Fu et al., 2026).

Emergent Behaviors of On-Policy Training. Recent studies have investigated intrinsic properties of on-policy RL training. Yue et al. (2025) find that RL primarily improves pass@1 sampling efficiency without expanding the model’s reasoning frontier. Cui et al. (2025b) identify early-stage entropy collapse that prematurely degrades exploration. Tan et al. (2026) reveal a power-law relationship between test loss, computation, and data volume, showing larger models exhibit superior learning efficiency. From a parameter-dynamics perspective, Cai et al. (2025) uncover Rank-1 dominance and Rank-1 linear dynamics, enabling $2.5\times$ acceleration via AlphaRL. Cai et al. (2026) attribute on-policy distillation’s efficiency to modular redundancy suppression and early directional stabilization, proposing EffOPD with $3\times$ speedup (Song & Zheng, 2026).

Steering Vectors and Representation Engineering. While most adaptation methods rely on parameter updates (full fine-tuning or LoRA), an alternative line of work intervenes directly on internal activations via vector steering. Representation Engineering (RepE) provides a unifying framework, treating population-level representations as the analytical unit for monitoring and manipulating cognitive phenomena (Zou et al., 2023). High-level behaviors often exhibit low-dimensional structure: refusal is mediated by a single direction (Arditi et al., 2024) or multi-dimensional concept cones (Wollschläger et al., 2025). Task vectors (Ilharco et al., 2022) and function vectors (Todd et al., 2024) enable behavioral editing through vector arithmetic, and steering is increasingly viewed as a paradigm complementary to fine-tuning and prompting (Ostermann et al., 2026).

B PRELIMINARIES AND EXPERIMENTAL SETUP

B.1 PRELIMINARIES

We consider three post-training paradigms: Reinforcement Learning (RL), On-Policy Distillation (OPD), and Off-Policy Distillation. Let π_θ denote the policy or student model, π^* the fixed teacher model, and π_{ref} the fixed reference model used for regularization. The teacher provides distillation targets, whereas the reference model anchors RL regularization; these roles are distinct.

Reinforcement Learning (RL). The RL objective is:

$$\max_{\theta} J_{\text{RL}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} [\mathbb{E}_{y \sim \pi_\theta(\cdot | x)} [r(x, y)] - \beta D_{\text{KL}}(\pi_\theta(\cdot | x) \parallel \pi_{\text{ref}}(\cdot | x))], \quad (9)$$

where $y = (y_1, \dots, y_T)$ is sampled from the current policy, $r(x, y)$ is the reward assigned to response y for query x , and β controls the strength of KL regularization.

In Reinforcement Learning from Verifiable Rewards (RLVR), $r(x, y)$ is determined by verifiable outcomes, such as answer correctness or unit-test execution, without requiring a learned reward model.

Separating the task-reward term from reference-policy regularization gives the policy-gradient form:

$$\begin{aligned} \nabla_{\theta} J_{\text{RL}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot | x)} & \left[\sum_{t=1}^T A_t^r \nabla_{\theta} \log \pi_\theta(y_t | x, y_{<t}) \right] \\ & - \beta \nabla_{\theta} \mathbb{E}_{x \sim \mathcal{D}} [D_{\text{KL}}(\pi_\theta(\cdot | x) \parallel \pi_{\text{ref}}(\cdot | x))], \end{aligned} \quad (10)$$

where A_t^r denotes the token-level advantage associated with the task reward. In RLVR, this reward is typically observed only after the complete response has been generated. Equation 10 describes the underlying regularized objective; practical GRPO and DAPO updates use algorithm-specific advantage estimates and clipped surrogate objectives.

On-Policy Distillation (OPD). OPD samples trajectories from the student policy and minimizes the reverse KL divergence from the student to the teacher:

$$\begin{aligned} \min_{\theta} J_{\text{OPD}}(\theta) &= \mathbb{E}_{x \sim \mathcal{D}} [D_{\text{KL}}(\pi_\theta(\cdot | x) \parallel \pi^*(\cdot | x))] \\ &= \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot | x)} \left[\sum_{t=1}^T (\log \pi_\theta(y_t | x, y_{<t}) - \log \pi^*(y_t | x, y_{<t})) \right]. \end{aligned} \quad (11)$$

The corresponding sequence-level gradient can be written as:

$$\begin{aligned} \nabla_{\theta} J_{\text{OPD}}(\theta) &= \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot | x)} \left[\sum_{t=1}^T \sum_{t'=t}^T (\log \pi_\theta(y_{t'} | x, y_{<t'}) - \log \pi^*(y_{t'} | x, y_{<t'})) \right. \\ &\quad \left. \cdot \nabla_{\theta} \log \pi_\theta(y_t | x, y_{<t}) \right]. \end{aligned} \quad (12)$$

For optimization, we use an immediate token-level approximation that retains only the contribution at $t' = t$:

$$\begin{aligned} \nabla_{\theta} J_{\text{OPD}}(\theta) &\approx \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot | x)} \left[\sum_{t=1}^T (\log \pi_\theta(y_t | x, y_{<t}) - \log \pi^*(y_t | x, y_{<t})) \right. \\ &\quad \left. \cdot \nabla_{\theta} \log \pi_\theta(y_t | x, y_{<t}) \right]. \end{aligned} \quad (13)$$

This approximation provides token-level supervision along student-generated trajectories, but omits the future-token contributions in Equation 12. When implemented as an advantage-weighted surrogate, the log-probability difference is treated as a stop-gradient weight, with gradients taken through the student log-probability that it multiplies.

Off-Policy Distillation (Off-PD). Off-policy distillation instead samples trajectories from the teacher policy. Its objective is the forward KL divergence from the teacher to the student:

$$\min_{\theta} J_{\text{Off-PD}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} [D_{\text{KL}}(\pi^*(\cdot | x) \parallel \pi_\theta(\cdot | x))]$$

$$= \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi^*(\cdot|x)} \left[\sum_{t=1}^T (\log \pi^*(y_t | x, y_{<t}) - \log \pi_\theta(y_t | x, y_{<t})) \right]. \quad (14)$$

Since the teacher-dependent term is constant with respect to θ , this objective is equivalent to minimizing the negative log-likelihood of teacher-generated trajectories:

$$\mathcal{L}_{\text{Off-PD}}(\theta) = -\mathbb{E}_{x \sim \mathcal{D}, y \sim \pi^*(\cdot|x)} \left[\sum_{t=1}^T \log \pi_\theta(y_t | x, y_{<t}) \right]. \quad (15)$$

The corresponding gradient is:

$$\nabla_\theta \mathcal{L}_{\text{Off-PD}}(\theta) = -\mathbb{E}_{x \sim \mathcal{D}, y \sim \pi^*(\cdot|x)} \left[\sum_{t=1}^T \nabla_\theta \log \pi_\theta(y_t | x, y_{<t}) \right]. \quad (16)$$

Thus, OPD and off-policy distillation differ in both trajectory sampling and the objective being optimized. OPD uses student-generated trajectories with a reverse-KL objective, whereas off-policy distillation uses teacher-generated trajectories with a forward-KL objective, implemented as sequence negative log-likelihood. The latter trains the student on teacher-generated prefixes, while the former evaluates the teacher on prefixes generated by the student.

B.2 EXPERIMENTAL SETUP

Overview. Each experiment in this paper is defined by a training paradigm and an update configuration. The three paradigms are on-policy RL with verifiable rewards (Eqn. 9), on-policy distillation (Eqn. 11), and off-policy distillation (Eqn. 15), as formalized in Section B.1. For RL, we consider full fine-tuning and LoRA, whereas for both on-policy and off-policy distillation, we additionally evaluate vector steering. These configurations differ in which parameters receive gradients and, consequently, in the dimensionality and expressiveness of the resulting updates. Within each matched comparison, we hold the base model, prompt set, teacher model when applicable, sampling recipe, and evaluation protocol fixed across the applicable update configurations. Parameterization-specific optimization settings are reported in Table 2, and execution differences are described below. In on-policy experiments, matching the sampling recipe does not imply identical trajectories, since the samples depend on the current policy. This setup allows us to compare update parameterizations under matched modeling and evaluation protocols while making their optimization differences explicit.

Models. To ensure the generality of our findings, we conduct experiments across model scales ranging from 1.5B to 14B parameters, spanning two model families: Qwen2.5/DeepSeek-R1-Distill (1.5B and 7B) and Qwen3 (4B, 8B, and 14B), and two RL algorithms (GRPO and DAPO). All RL models are trained locally using the `verl` framework (Sheng et al., 2025). For all distillation students, the default teacher is the RL-tuned version of the student’s own base model, ensuring that the teacher–student gap corresponds exactly to the RL update itself.

Datasets and Evaluation. The training data spans four topics, each with its own verifier and held-out evaluation set (Table 1). Rewards are always *verifiable* in the sense of §B.1: rule-based answer matching for mathematical and scientific reasoning, unit-test execution for code, and programmatic constraint checking for instruction following. For the off-policy distillation experiments in Figure 2 (d), we additionally compare two sources of supervision: standard SFT on Math-CoT-44K (Ren et al., 2026), which contains approximately 44K mathematical queries and 32 responses sampled from Qwen3-32B for each query, yielding more than 1.4M reasoning traces; and off-policy distillation on OpenMathReasoning (Moshkov et al., 2025), which contains 540K mathematical problems and 3.2M long chain-of-thought solutions generated by DeepSeek-R1 and QwQ-32B.

Table 1: Training and evaluation datasets and reward functions for each domain.

Domain	Training Set	Evaluation Set	Reward
Math	DeepMath-103K (He et al., 2025)	AIME 2024 (Ye et al., 2025) MATH500 (Lightman et al., 2023)	Answer matching
Science	SciKnowEval-Training (Feng et al., 2024)	SciKnowEval-Eval (Feng et al., 2024)	Answer matching
Code	Eurus (Cui et al., 2025a)	LiveCodeBench-v5 (Naman Jain, 2024)	Unit tests
Instruction following	IFEval-Training (Zhao et al., 2024)	IFEval-Eval (Zhao et al., 2024)	Constraint checking

Instruction-following split. For instruction following, we partition the dataset at the prompt level into an 80% training split and a 20% held-out evaluation split, denoted as IFEval-Training and IFEval-Eval, respectively. These names refer to the partitions used in this study. All responses associated with the same prompt remain in the same partition, and evaluation prompts are excluded from training. The split is kept fixed across all compared configurations.

Evaluation protocol. We report `mean@4` on the held-out evaluation sets. For each evaluation prompt, we independently sample four responses using the same decoding configuration. Each response is scored using the corresponding task-specific evaluation criterion. The reported result is the arithmetic mean of the four single-sample evaluation scores computed on the same held-out set:

$$\text{mean@4} = \frac{1}{4} \sum_{j=1}^4 \text{Score} \left(\{(x_i, y_i^{(j)})\}_{i=1}^N \right), \quad (17)$$

where N is the number of evaluation prompts, $y_i^{(j)}$ is the j -th sampled response to prompt x_i , and `Score` is the task-specific evaluation metric. For mean per-response correctness, this is equivalently the average over all $4N$ scored responses. We do not select the best of the four responses or apply majority voting; `mean@4` is therefore distinct from `pass@4`.

Update configurations. We compare the following trainable parameterizations under the matched modeling and evaluation protocols described above, using the configuration-specific learning rates in Table 2.

- **Full fine-tuning** updates all model parameters by default and serves as the unrestricted reference configuration. In principle, it can express a substantially broader class of parameter updates than the other configurations.
- **LoRA** (Hu et al., 2021) injects low-rank adapters on `all-linear` target modules while keeping the backbone frozen. This restricts the update to a low-rank form, though each input can still produce a different correction (i.e., it remains context-dependent). The rank varies across experiments; Table 2 reports the default rank-8 configuration.
- **Vector steering** freezes the entire backbone and trains context-invariant bias vectors added to the residual stream. For a controlled layer set $\mathcal{S}$, the single-vector intervention is

$$\mathbf{h}_\ell \mapsto \mathbf{h}_\ell + \mathbf{v}_\ell, \quad \mathbf{v}_\ell \in \mathbb{R}^d, \quad \ell \in \mathcal{S}, \quad (18)$$

with $|\mathcal{S}|d$ trainable parameters. For the default single-vector configuration on Qwen3-4B ($d = 2560$) with $\mathcal{S} = \{5, \dots, 20\}$ and $|\mathcal{S}| = 16$, this amounts to 40,960 scalars, approximately 4.1×10^4 , or roughly 1.0×10^{-5} of the backbone. Crucially, $\mathbf{v}_\ell$ does not vary with the input: the *same* vector is added regardless of context or decoding position.

Unless otherwise stated, vector steering uses the controlled layer set $\mathcal{S} = \{5, \dots, 20\}$ and a single vector per controlled layer ($K = 1$). In the layer-wise experiments of Section 2.3, all three configurations are restricted to the selected layer: full-parameter updating trains only that layer, LoRA adapts its linear modules, and vector steering adds an offset to its output. The remainder of the backbone is frozen. These selected-layer experiments are distinct from the default model-wide full-fine-tuning and LoRA configurations.

For multi-vector probes ($K > 1$), we learn the ordered set $\{\mathbf{v}_\ell^{(0)}, \dots, \mathbf{v}_\ell^{(K-1)}\}$ sequentially at each controlled layer. The first vector is trained to convergence and then retained as a fixed reference. At each subsequent stage k , only the current vector $\mathbf{v}_\ell^{(k)}$ is injected and optimized; previously learned vectors remain inactive in the forward pass and are used solely to define the orthogonality constraint. After each optimizer step, the current vector is reprojected onto the orthogonal complement of the previously learned directions. Training advances to the next vector once the current-stage loss plateaus. Each direction is also evaluated independently, with only the corresponding vector active at each controlled layer. Consequently, each stage optimizes $|\mathcal{S}|d$ parameters, while the complete collection contains $K|\mathcal{S}|d$ scalars; increasing K probes additional independently effective directions rather than simultaneously summing constant offsets.

Vector steering requires a substantially larger learning rate than the other two configurations; the values we use are reported in Table 2.

The auxiliary reference-KL coefficient in Table 2 refers to additional regularization toward π_{ref} , not to the main distillation objective. In particular, a zero auxiliary coefficient does not remove the reverse-KL teacher supervision in OPD or the teacher-trajectory negative log-likelihood in Off-PD.

Prompts. All Qwen3 models are run and trained in *non-thinking* mode (`enable_thinking=False` passed to the chat template) so that the response distribution is directly comparable between the base model, the RL teacher, and every steered student; DeepSeek-R1-Distill students keep thinking mode enabled, as their base model is trained that way. We always use each model’s own built-in chat template rather than a hand-written wrapper, and do not manually prepend an additional system message. Each domain uses its own answer-format instruction, which is also what the corresponding verifier parses. The examples below show *user-message content only*, with `{question}` denoting the raw problem statement. Role delimiters and model-specific thinking prefixes are supplied by the native chat template rather than manually inserted into the user message.

For **mathematical reasoning**, the answer-extraction instruction is appended to the question:

Table 2: Learning rates and paradigm-specific hyperparameters.

	RL	OPD	Off-PD
LR, full fine-tuning	1×10^{-6}	1×10^{-5}	5×10^{-5}
LR, LoRA ($r=8$)	3×10^{-5}	5×10^{-6}	1×10^{-5}
LR, vector steering	—	1×10^{-1}	5×10^{-2}
Max prompt / response	2,048 / 20,480	3,072 / 16,384	3,072 / 16,384
Prompt batch	128	1,024	1,024
Mini-batch (per step)	32	1,024	1,024
Training samples per prompt n	16	1	1
Evaluation samples per prompt	4	4	4
Evaluation metric	mean@4	mean@4	mean@4
Rollout source	student	student	frozen teacher
Rollout IS correction	—	token, threshold 5.0	off
Auxiliary reference-KL coeff.	0.001 (GRPO) / 0 (DAPO)	0	0
Epochs	1000	1000	1000
Precision	bfloat16	bfloat16	bfloat16

{question}
Please reason step by step, and put your final answer within \boxed{ }.

For **scientific reasoning**, the instruction is placed first and the question is a multiple-choice item whose options are enumerated A–J, so that the verifier can match a single letter:

Solve the following problem. Make sure to put the answer (and only answer) inside \boxed{ }.

{question}
A: {option_A}
B: {option_B}
...
J: {option_J}

For **code generation**, the instruction requires a single fenced Python block, which the sandbox extracts verbatim and executes against the hidden unit tests:

{question}

Write Python code to solve the problem. Present the code in
```python  
Your code  
```  
at the end.
You need to think first then write the Python code.

For **instruction following**, the constraints are stated in natural language at the head of the prompt and no separate reasoning instruction is added, since the verifiable constraints concern the response’s surface form:

Please follow these instructions in your response: {constraint_1} and {constraint_2} and ... {constraint_n}. {question}

Teacher rollouts for off-policy distillation are generated with the *same* model-specific prompt template as the corresponding on-policy runs. Prompt construction is therefore matched, while the two paradigms retain the distinct trajectory-sampling distributions and distillation objectives defined in Section B.1.

Infrastructure. All training runs use $8\times$ or $32\times$ H20 96GB GPUs with FSDP for the actor and vLLM for generation. Vector-steering runs disable CUDA graph capture (`enforce_eager=true`) because the steering hook mutates hidden states inside the vLLM forward pass; LoRA and full-parameter runs leave CUDA graphs enabled. Rollout tensor parallelism is 4 for RL and 2 for distillation. When a fixed model is used for scoring, its parameters are offloaded to CPU between forward passes; this model serves as the reference policy for RL regularization or the teacher for distillation supervision, depending on the paradigm. The two commands below give the on-policy and off-policy distillation invocations for vector steering. The full-parameter and LoRA variants

use the corresponding update parameterizations and hyperparameters in Table 2, with the configuration-specific generation-engine settings described above.

On-Policy Distillation Command (vector steering)

```
# examples/representation/4bopd_single.sh -> 4bopd.sh
python3 -m verl.trainer.main_ppo \
  algorithm.adv_estimator=grpo \
  algorithm.use_kl_in_reward=False \
  algorithm.rollout_correction.rollout_is=token \
  algorithm.rollout_correction.rollout_is_threshold=5.0 \
  algorithm.rollout_correction.rollout_rs=null \
  algorithm.rollout_correction.bypass_mode=false \
  data.train_files=/path/to/DeepMath-103K/train_filtered_level6.parquet \
  data.val_files=["/path/to/AIME2024/test.parquet"] \
  data.train_batch_size=1024 \
  data.max_prompt_length=3072 \
  data.max_response_length=16384 \
  data.filter_overlong_prompts=True \
  data.truncation='error' \
  data.shuffle=True \
  data.seed=42 \
  data.return_raw_chat=True \
  +data.apply_chat_template_kwargs.enable_thinking=False \
  actor_rollout_ref.model.path=$MODEL_PATH \
  +actor_rollout_ref.model.base_model_path=$MODEL_PATH \
  +actor_rollout_ref.ref.model.path=$TEACHER_MODEL_PATH \
  actor_rollout_ref.model.lora_rank=0 \
  actor_rollout_ref.model.enable_trainable_token_vector=true \
  actor_rollout_ref.model.trainable_token_vector_mode=single \
  actor_rollout_ref.model.trainable_token_vector_num=1 \
  actor_rollout_ref.model.trainable_token_vector_layer_start=5 \
  actor_rollout_ref.model.trainable_token_vector_layer_end=20 \
  actor_rollout_ref.model.trainable_token_vector_sampling_method=hypersphere \
  actor_rollout_ref.model.trainable_token_vector_scale=0.1 \
  actor_rollout_ref.model.trainable_token_vector_learnable_alpha=false \
  actor_rollout_ref.model.trainable_token_vector_alpha_init=0.0 \
  actor_rollout_ref.model.trainable_token_vector_curriculum=none \
  actor_rollout_ref.model.trainable_token_vector_force_all_tokens=true \
  actor_rollout_ref.model.use_remove_padding=True \
  actor_rollout_ref.model.enable_gradient_checkpointing=True \
  actor_rollout_ref.actor.optim.lr=1e-1 \
  actor_rollout_ref.actor.optim.lr_warmup_steps_ratio=0.0 \
  actor_rollout_ref.actor.policy_loss.only_reverse_kl_advantages=True \
  actor_rollout_ref.actor.ppo_mini_batch_size=1024 \
  actor_rollout_ref.actor.ppo_micro_batch_size_per_gpu=1 \
  actor_rollout_ref.actor.use_kl_loss=True \
  actor_rollout_ref.actor.kl_loss_coef=0 \
  actor_rollout_ref.actor.kl_loss_type=low_var_kl \
  actor_rollout_ref.actor.entropy_coeff=0 \
  actor_rollout_ref.actor.ppo_max_token_len_per_gpu=32768 \
  actor_rollout_ref.actor.fsdp_config.dtype=bfloat16 \
  actor_rollout_ref.actor.fsdp_config.model_dtype=fp32 \
  actor_rollout_ref.actor.fsdp_config.use_orig_params=True \
  actor_rollout_ref.rollout.name=vllm \
  actor_rollout_ref.rollout.calculate_log_probs=true \
  actor_rollout_ref.rollout.n=1 \
  actor_rollout_ref.rollout.temperature=1.0 \
  actor_rollout_ref.rollout.top_p=1.0 \
  actor_rollout_ref.rollout.tensor_model_parallel_size=2 \
  actor_rollout_ref.rollout.gpu_memory_utilization=0.8 \
  actor_rollout_ref.rollout.enforce_eager=true \
  actor_rollout_ref.rollout.free_cache_engine=True \
  actor_rollout_ref.rollout.max_num_batched_tokens=32768 \
  actor_rollout_ref.rollout.val_kwargs.do_sample=True \
  actor_rollout_ref.rollout.val_kwargs.temperature=1.0 \
  actor_rollout_ref.rollout.val_kwargs.top_p=1.0 \
  actor_rollout_ref.rollout.val_kwargs.n=4 \
  actor_rollout_ref.ref.log_prob_micro_batch_size_per_gpu=1 \
```

```

actor_rollout_ref.ref.fsdg_config.param_offload=True \
reward_model.reward_manager=naive \
trainer.save_vector=true \
trainer.save_vector_dir=/path/to/trainable_vectors \
trainer.logger=['console',"wandb"] \
trainer.n_gpus_per_node=8 \
trainer.nnodes=1 \
trainer.test_freq=10 \
trainer.total_epochs=1000 $@

```

Off-Policy Distillation Command (vector steering)

```

# Stage 1: dump teacher rollouts once
python3 -m verl.trainer.main_generation \
    model.path=$TEACHER_MODEL_PATH \
    data.path=/path/to/DeepMath-103K/train_filtered_level6.parquet \
    data.prompt_key=prompt \
    data.n_samples=1 \
    data.batch_size=1024 \
    rollout.temperature=1.0 \
    rollout.top_p=1.0 \
    rollout.prompt_length=3072 \
    rollout.response_length=16384 \
    rollout.max_num_batched_tokens=32768 \
    rollout.tensor_model_parallel_size=2 \
    data.output_path=$OUTPUT_BASE/round_${i}.parquet

python3 examples/representation/merge_rollouts_to_sft.py \
    --inputs $OUTPUT_BASE/round_*.parquet \
    --output $OUTPUT_BASE/teacher_sft_all.parquet \
    --prompt_key prompt --response_key response

# Stage 2: distill off-policy on the frozen corpus
python3 -m verl.trainer.main_ppo \
    algorithm.adv_estimator=grpo \
    algorithm.rollout_correction.rollout_is=null \
    algorithm.use_kl_in_reward=False \
    data.train_files=$OUTPUT_BASE/teacher_sft_all.parquet \
    data.val_files=["'/path/to/AIME2024/test.parquet'"] \
    data.train_batch_size=1024 \
    data.max_prompt_length=3072 \
    data.max_response_length=16384 \
    data.return_raw_chat=True \
    +data.apply_chat_template_kwargs.enable_thinking=False \
    actor_rollout_ref.model.path=$MODEL_PATH \
    +actor_rollout_ref.ref.model.path=$TEACHER_MODEL_PATH \
    actor_rollout_ref.model.lora_rank=0 \
    actor_rollout_ref.model.enable_trainable_token_vector=true \
    actor_rollout_ref.model.trainable_token_vector_mode=single \
    actor_rollout_ref.model.trainable_token_vector_num=1 \
    actor_rollout_ref.model.trainable_token_vector_layer_start=5 \
    actor_rollout_ref.model.trainable_token_vector_layer_end=20 \
    actor_rollout_ref.model.trainable_token_vector_sampling_method=hypersphere \
    actor_rollout_ref.model.trainable_token_vector_scale=0.1 \
    actor_rollout_ref.model.trainable_token_vector_learnable_alpha=false \
    actor_rollout_ref.model.trainable_token_vector_alpha_init=0.0 \
    actor_rollout_ref.model.trainable_token_vector_curriculum=none \
    actor_rollout_ref.model.trainable_token_vector_force_all_tokens=true \
    actor_rollout_ref.actor.optim.lr=5e-2 \
    actor_rollout_ref.actor.policy_loss.only_reverse_kl_advantages=true \
    actor_rollout_ref.actor.ppo_mini_batch_size=1024 \
    actor_rollout_ref.actor.ppo_micro_batch_size_per_gpu=1 \
    actor_rollout_ref.actor.ppo_max_token_len_per_gpu=32768 \
    actor_rollout_ref.actor.use_kl_loss=True \
    actor_rollout_ref.actor.kl_loss_coef=0 \
    actor_rollout_ref.actor.entropy_coef=0 \
    actor_rollout_ref.rollout.offline_teacher_rollout=true \
    actor_rollout_ref.rollout.offline_teacher_data_path=./teacher_sft_all.parquet \
    actor_rollout_ref.rollout.offline_teacher_response_key=response \

```

```
actor_rollout_ref.rollout.name=vllm \  
actor_rollout_ref.rollout.n=1 \  
actor_rollout_ref.rollout.tensor_model_parallel_size=2 \  
actor_rollout_ref.rollout.gpu_memory_utilization=0.5 \  
actor_rollout_ref.rollout.enforce_eager=true \  
actor_rollout_ref.ref.fsdp_config.param_offload=True \  
reward_model.reward_manager=naive \  
trainer.save_vector=true \  
trainer.logger=['"console"', '"wandb"'] \  
trainer.n_gpus_per_node=8 \  
trainer.nnodes=1 \  
trainer.test_freq=5 \  
trainer.total_epochs=1000 $@
```

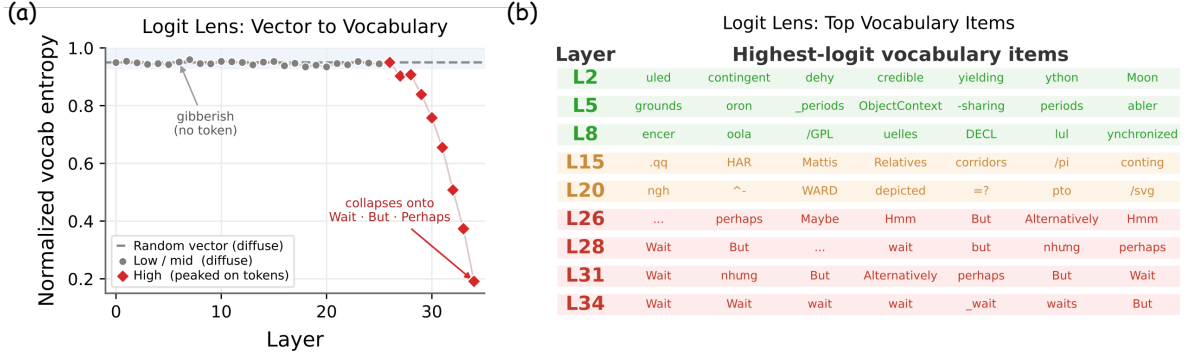


Figure 9: Vocabulary-space readout of steering vectors across network depth. (a) Normalized entropy of $\text{softmax}(\text{RMSNorm}(\mathbf{v}_L)\mathbf{W}_U^\top)$. The dashed line at 0.9498 denotes the mean entropy of normalized Gaussian directions passed through the same readout pipeline. (b) Highest-logit vocabulary items for steering vectors learned at representative layers. All vectors are normalized before readout.

C ADDITIONAL EXPERIMENTS FOR EFFECTIVE MANIFOLD CAPACITY

C.1 VOCABULARY-SPACE READOUT

The layer-wise experiments in the main text show that fixed-vector steering recovers most of the RL-induced performance gain at lower and intermediate layers, but becomes substantially less effective near the output. To better understand this difference, we examine the vocabulary-space readout of steering vectors learned at different depths. This analysis characterizes how the learned directions align with output vocabulary items and how this alignment changes with injection depth.

For a steering vector $\mathbf{v}_L \in \mathbb{R}^H$ learned at layer L , we bypass subsequent Transformer layers and directly apply RMS normalization followed by the unembedding matrix (Geva et al., 2022):

$$\ell(\mathbf{v}_L) = \mathbf{W}_U \text{RMSNorm}(\mathbf{v}_L), \quad \text{RMSNorm}(\mathbf{v}) = \frac{\mathbf{v}}{\sqrt{\frac{1}{H} \sum_{j=1}^H v_j^2 + \varepsilon}}, \quad \varepsilon = 10^{-6}, \quad (19)$$

where $\mathbf{W}_U \in \mathbb{R}^{V \times H}$ is the unembedding matrix, and H and V denote the hidden dimension and vocabulary size, respectively. The resulting readout distribution is

$$\mathbf{p}_L = \text{softmax}(\ell(\mathbf{v}_L)). \quad (20)$$

The learned vector norms vary across injection depths. RMS normalization reduces the influence of these magnitude differences, allowing the readout to primarily characterize the vocabulary alignment of each steering direction.

We quantify the concentration of the readout distribution using normalized entropy:

$$\tilde{\mathcal{H}}(\mathbf{p}_L) = -\frac{1}{\log V} \sum_{t=1}^V p_{L,t} \log p_{L,t}. \quad (21)$$

Values closer to one indicate a more diffuse distribution, whereas lower values indicate greater concentration on a smaller set of vocabulary items.

To establish a random-direction baseline, we sample isotropic Gaussian vectors with the same dimensionality as the steering vectors, normalize them to unit L_2 norm, and pass them through the same RMS normalization, unembedding, and softmax operations:

$$\mathbf{z}^{(m)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_H), \quad \bar{\mathbf{z}}^{(m)} = \frac{\mathbf{z}^{(m)}}{\|\mathbf{z}^{(m)}\|_2}, \quad (22)$$

$$\mathbf{p}_{\text{null}}^{(m)} = \text{softmax}(\mathbf{W}_U \text{RMSNorm}(\bar{\mathbf{z}}^{(m)})). \quad (23)$$

Averaging the normalized entropy over the M sampled directions gives

$$\tilde{\mathcal{H}}_{\text{null}} = \frac{1}{M} \sum_{m=1}^M \tilde{\mathcal{H}}(\mathbf{p}_{\text{null}}^{(m)}) = 0.9498. \quad (24)$$

As shown in Figure 9(a), the normalized entropy remains close to the random baseline at lower and intermediate layers, indicating a diffuse vocabulary readout. The highest-logit items from L2 to L20 in Figure 9(b) likewise

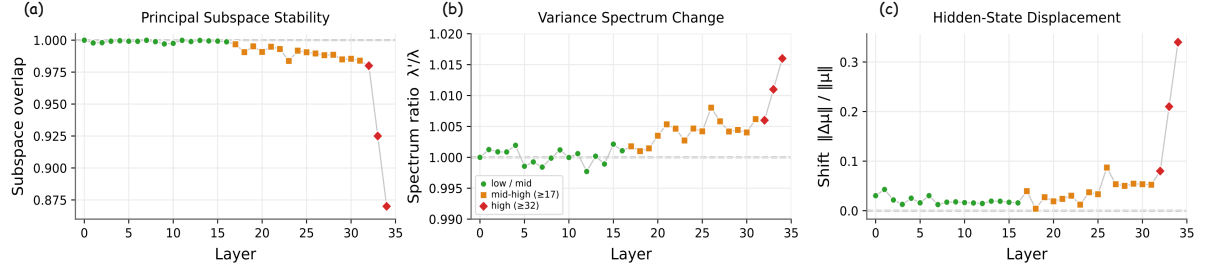


Figure 10: Effect of single-vector steering on hidden-state geometry in Qwen3-4B ($k = 20$, $H = 2560$), measured at layer $L + 1$ for injection at layer L . (a) Mean principal-subspace overlap between clean and steered activations. (b) Ratio of the top- k explained variance after and before steering. (c) Relative displacement of the activation centroid. Dashed lines denote the corresponding reference values.

consist mainly of heterogeneous subword fragments and symbols, without an apparent common semantic theme. Combined with the strong performance of steering at these depths, this observation shows that effective intermediate-layer interventions need not exhibit a concentrated direct preference for particular output tokens. This pattern is consistent with the role of downstream computation: as a shared additive correction propagates through the remaining layers, its interaction with different hidden states can produce context-dependent output adjustments.

Near the output, the readout becomes increasingly concentrated. Beginning around L26, normalized entropy falls below the random baseline, while the highest-logit items increasingly include *Wait*, *But*, *perhaps*, and *Alternatively*. These tokens are associated with contrast, qualification, and reconsideration, suggesting that high-layer steering directions have more explicit vocabulary-level associations. The readout thus shifts from a diffuse pattern at lower and intermediate layers toward a concentrated set of discourse-related items near the output.

This increase in vocabulary interpretability does not coincide with better steering performance. Instead, the main-text depth sweep shows that fixed-vector steering deteriorates at these higher layers. A direction can therefore have a clear vocabulary-level association while remaining insufficient for effective task adaptation. The issue is not simply whether a direction aligns with meaningful output tokens, but how that direction is applied across token positions. Expressions of contrast, qualification, or reconsideration are useful in different contexts, whereas a fixed steering vector applies the same additive correction at every position, without explicitly adjusting its strength to the current representation.

Summary. The vocabulary readout provides a complementary view of the layer-wise results. Lower- and intermediate-layer steering remains effective despite its diffuse direct readout, whereas high-layer directions exhibit more concentrated vocabulary associations alongside weaker fixed-vector performance. This pattern suggests that direct vocabulary alignment alone does not account for steering effectiveness. One possible explanation is that the remaining layers allow earlier interventions to acquire context-dependent effects, while later interventions have less opportunity for such transformation. The gains from high-layer gating are consistent with this account, supporting input dependence as a useful factor in understanding the depth sensitivity of fixed-vector steering.

C.2 LOCAL REPRESENTATION GEOMETRY UNDER SINGLE-VECTOR STEERING

Section 2.3 shows that single-vector steering recovers most of the RL-induced performance gain at lower and intermediate layers, but becomes less effective near the output. The training dynamics and gating results suggest that the expressiveness of a fixed additive correction contributes to this depth dependence. Here, we complement the performance analysis by examining changes in hidden-state geometry across injection depths, focusing on the leading principal subspaces, their associated variance, and the activation means.

Experimental Setup. For each injection layer L , we collect hidden states from the base and steered models on the same held-out inputs. The steering vector is added to the output of layer L , and representations are measured at the output of layer $L + 1$. This captures the local effect of steering after one subsequent Transformer layer. Aggregating all valid token positions gives

$$\mathbf{X}_L \in \mathbb{R}^{N \times H}, \quad \mathbf{X}'_L \in \mathbb{R}^{N \times H}, \quad (25)$$

where N is the number of token positions and H is the hidden dimension. The subscript L indexes the injection layer; both matrices contain activations measured at layer $L + 1$. Their respective means are

$$\mu_L = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_{L,i}, \quad \mu'_L = \frac{1}{N} \sum_{i=1}^N \mathbf{x}'_{L,i}. \quad (26)$$

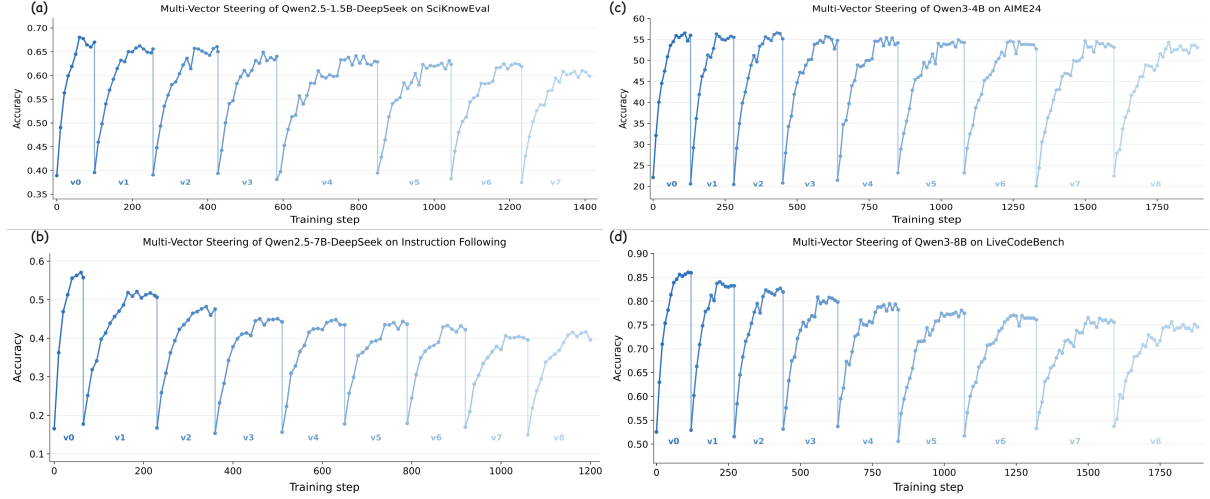


Figure 11: Multi-vector steering across all four task domains and model scales. The first vector is consistently the most effective, with subsequent directions showing gradually decreasing standalone performance; the decay is slower for larger models.

We center each activation matrix using its own mean and independently compute low-rank singular value decompositions:

$$\mathbf{X}_{L,c} \approx \mathbf{U}_L \Sigma_L \mathbf{V}_L^\top, \quad \mathbf{X}'_{L,c} \approx \mathbf{U}'_L \Sigma'_L \mathbf{V}'_L{}^\top. \quad (27)$$

The first k right singular vectors form the orthonormal principal-component bases $\mathbf{V}_{L,k}, \mathbf{V}'_{L,k} \in \mathbb{R}^{H \times k}$. Writing $\Sigma_{L,j}$ and $\Sigma'_{L,j}$ for the corresponding singular values, the covariance eigenvalues are

$$\lambda_{L,j} = \frac{\Sigma_{L,j}^2}{N-1}, \quad \lambda'_{L,j} = \frac{\Sigma'_{L,j}{}^2}{N-1}. \quad (28)$$

For Qwen3-4B, we use $k = 20$ and $H = 2560$.

Principal-Subspace Alignment. We measure the alignment between the clean and steered top- k principal subspaces as

$$\text{Overlap}_{L,k} = \frac{1}{k} \sum_{j=1}^k \sigma_j(\mathbf{V}_{L,k}^\top \mathbf{V}'_{L,k}), \quad (29)$$

where $\sigma_j(\cdot)$ denotes the j -th singular value. These singular values are the cosines of the principal angles between the two subspaces, so an overlap close to one indicates similar subspace orientations.

As shown in Figure 10(a), the overlap remains close to one for lower- and intermediate-layer injections. At these depths, the leading directions of activation variation therefore remain largely aligned after one layer of propagation. The overlap decreases toward the output, with the largest departures occurring at the latest injection sites. This indicates greater changes in the orientation of the leading subspace for these high-layer interventions.

Variance Captured by the Leading Components. We next compare the total variance captured by the top- k components of the steered and clean activations:

$$\rho_{L,k} = \frac{\sum_{j=1}^k \lambda'_{L,j}}{\sum_{j=1}^k \lambda_{L,j}}. \quad (30)$$

The numerator and denominator use the leading eigenvalues of their respective covariance matrices. A ratio close to one indicates similar amounts of variance captured by the two principal subspaces.

Figure 10(b) shows that this ratio stays close to one at lower and intermediate injection depths. Along with the high subspace overlap, this suggests that steering at these depths introduces relatively small changes in both the orientation and the total variance of the leading components. The ratio departs further from one near the output, although the changes in captured variance remain modest. Thus, the reduced subspace alignment at the latest injection sites is accompanied by a comparatively small change in the variance captured by the leading components.

Displacement of the Activation Centroid. To characterize changes in the mean representation, we measure the relative centroid displacement:

$$D_L = \frac{\|\mu'_L - \mu_L\|_2}{\|\mu_L\|_2 + 10^{-9}}. \quad (31)$$

This statistic expresses the change in the activation mean relative to the norm of the clean mean.

As shown in Figure 10 (c), relative centroid displacement is small for lower- and intermediate-layer injections and increases near the output, particularly at the latest injection sites. The strong performance of earlier-layer steering therefore coincides with only a small relative change in the mean representation at the next layer. High-layer steering produces a larger relative mean shift, despite its weaker task performance.

Relation to the Layer-Wise Expressiveness Bottleneck. Effective steering at lower and intermediate layers is accompanied by limited changes in the measured local geometry. Near the output, reduced subspace alignment and larger centroid shifts coexist with weaker task performance. These observations are consistent with the expressiveness analysis in Section 2.3: high-layer degradation occurs despite measurable changes in the representations, and the gains from gating suggest that input-dependent modulation helps make these interventions more effective.

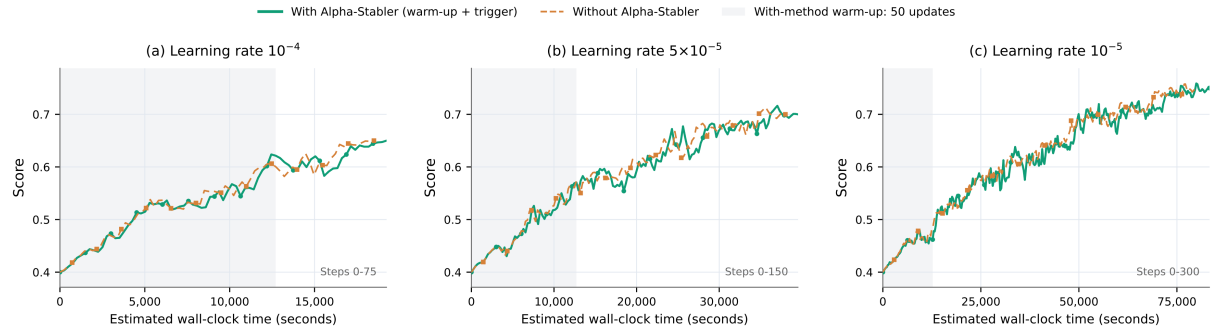


Figure 12: Illustrative comparison of pre-collapse score trajectories with and without Alpha-Stabler at learning rates of (a) 10^{-4} , (b) 5×10^{-5} , and (c) 10^{-5} . Shaded regions denote the first 50 warm-up updates of Alpha-Stabler. Controlled and uncontrolled runs exhibit essentially the same score growth before collapse; the controlled runs avoid the late-stage score drop caused by geometric drift, indicating that Alpha-Stabler’s stabilization introduces almost no additional computational overhead.

Algorithm 1 Alpha-Stabler: Warm-Up Calibration and PSI-Triggered Gradient Projection

Require: Actor θ ; frozen initial base model θ_0 ; monitored layers $\mathcal{S}$ at 1/4, 1/2, and 3/4 network depth
Require: Principal fraction $q = 0.10$; warm-up steps $T_{\text{warm}} = 50$; monitoring interval $M = 3$; EMA coefficient $\beta = 0.95$; persistence $p = 3$; thresholds $c_{\text{on}} = 3$, $c_{\text{safe}} = 2$
Require: Baseline RL optimizer and learning rate η_θ ; numerical stabilizer $\epsilon > 0$; mean-shift energy floor $\eta_{\min} > 0$
Ensure: Trained actor parameters θ (inference uses the actor alone)

- 1: Run T_{warm} ordinary RL optimizer updates without projection; accumulate base-model activation covariance statistics on actor-generated rollouts
- 2: Every M warm-up updates, retain detached activation-shift matrices on the same token sequences in $\mathcal{X}_\ell$ if $N > 0$ and $\|X_\ell\|_F^2/N > \eta_{\min}$
- 3: Estimate $U_{\ell,r}$ from the accumulated reference covariance, with $r = \lceil qd \rceil$; freeze all bases
- 4: **for** each $\ell \in \mathcal{S}$ **do**
- 5: $\mathcal{W}_\ell \leftarrow \{\|XU_{\ell,r}\|_F^2/(\|X\|_F^2 + \epsilon) : X \in \mathcal{X}_\ell\}$
- 6: Stop and recalibrate if $\mathcal{W}_\ell$ is empty or the thresholds below violate $0 < \tau_\ell^{\text{safe}} < \tau_\ell^{\text{on}} < 1$
- 7: $m_\ell \leftarrow \text{median}_{\psi \in \mathcal{W}_\ell} \psi$; $s_\ell \leftarrow 1.4826 \text{ median}_{\psi \in \mathcal{W}_\ell} |\psi - m_\ell| + \epsilon$
- 8: $\tau_\ell^{\text{on}} \leftarrow m_\ell + c_{\text{on}} s_\ell$; $\tau_\ell^{\text{safe}} \leftarrow m_\ell + c_{\text{safe}} s_\ell$
- 9: $a_\ell \leftarrow 0$; $n_\ell \leftarrow 0$; $\bar{\psi}_\ell \leftarrow m_\ell$
- 10: **end for**
- 11: Discard warm-up calibration records; $t \leftarrow T_{\text{warm}}$
- 12: **while** training is not complete **do**
- 13: Sample actor trajectories and compute rewards and advantages; store behavior log probabilities
- 14: **for** each optimization minibatch $\mathcal{B}$ corresponding to one optimizer update **do**
- 15: $t \leftarrow t + 1$; clear parameter gradients
- 16: Run the actor forward and compute $\mathcal{L}_{\text{RL}}$, retaining its computation graph
- 17: **if** $t \bmod M = 0$ **then**
- 18: Obtain frozen-base activations on the same token sequences without gradient tracking; $N \leftarrow$ number of valid token positions
- 19: **for** each $\ell \in \mathcal{S}$ **do**
- 20: $X_\ell \leftarrow \text{sg}(H_\ell(\theta; \mathcal{B}) - H_\ell(\theta_0; \mathcal{B}))$ over valid tokens
- 21: **if** $N > 0$ and $\|X_\ell\|_F^2/N > \eta_{\min}$ **then**
- 22: $\psi_\ell \leftarrow \|X_\ell U_{\ell,r}\|_F^2/(\|X_\ell\|_F^2 + \epsilon)$
- 23: $\bar{\psi}_\ell \leftarrow \beta \bar{\psi}_\ell + (1 - \beta) \psi_\ell$
- 24: $n_\ell \leftarrow n_\ell + 1$ if $\bar{\psi}_\ell > \tau_\ell^{\text{on}}$; otherwise $n_\ell \leftarrow 0$
- 25: Set $a_\ell \leftarrow 1$ if $n_\ell \geq p$; set $a_\ell \leftarrow 0$ if $\bar{\psi}_\ell < \tau_\ell^{\text{safe}}$; otherwise retain a_ℓ
- 26: **else**
- 27: $n_\ell \leftarrow 0$ $\triangleright$ Retain $\bar{\psi}_\ell$ and a_ℓ
- 28: **end if**
- 29: **end for**
- 30: **end if**
- 31: Hold all flags fixed during backpropagation and the optimizer step
- 32: Backpropagate $\mathcal{L}_{\text{RL}}$, replacing each incoming monitored gradient by $\tilde{G}_\ell = G_\ell - a_\ell(G_\ell U_{\ell,r})U_{\ell,r}^\top$
- 33: Update θ using the baseline optimizer at learning rate η_θ
- 34: **end for**
- 35: **end while**
- 36: **return** θ

D THEORY OF LOW-DIMENSIONAL RLVR STEERING

We develop a theoretical account of when RLVR-induced gains admit a compact activation-space description. The analysis proceeds in three parts. **Part I** formulates local distillation as a Fisher-weighted least-squares problem in Section D.2, identifies sufficient conditions for shared teacher-induced shifts in Section D.3, and derives a recovery identity and a three-constant lower bound in Section D.4. **Part II** characterizes constant-vector fitting under sequential orthogonality constraints in Section D.5 and relates effective directions to activation variance and output sensitivity in Section D.6. **Part III** analyzes Alpha-Stabler’s fixed-reference PSI diagnostic and backward gradient projection in Section D.7. Section D.8 collects the main results. Two relations organize the recovery analysis:

$$\rho = \frac{\text{SNR}}{1 + \text{SNR}} \quad \text{and} \quad \rho \geq \frac{\kappa}{(1 + \eta)(1 + \varepsilon)}.$$

Here ε bounds principal energy relative to complement energy, η measures residual energy relative to the shared complement component, and κ measures consistency of its amplitude. The quantity ρ denotes recovery of locally representable target energy in the quadratic model, as defined in Definition D.2.

INFORMAL PREVIEW

Distillation into a frozen student can be viewed as fitting a hidden-state correction to the teacher’s output distribution. Local logit and KL expansions turn this fit into a weighted least-squares problem with a context-dependent target $\Delta(x)$, as established in Proposition D.1. To identify a source of shared structure in these targets, Lemma D.2 expresses the KL-regularized RL optimum as an exponential reward tilt, and Lemma D.3 relates its next-token probabilities to continuation-success probabilities. Shared success profiles and consistent representation pullbacks then yield a shared correction with a controlled residual through Lemma D.4, Corollary D.1, and Lemma D.5. Under a constant metric, the optimal fixed vector is the mean target shift. Its recovery is high when this mean dominates the context-dependent fluctuations, as shown in Theorem D.1. Table 3 summarizes the quantities entering the lower bound.

Table 3: Geometric quantities governing the sufficient recovery bound.

Const.	Name	Meaning	Related analysis
ε	principal intrusion	Upper bound on principal-to-complement energy	PSI and principal-subspace intervention under compatible metrics
η	complement residual	Residual-to-shared energy within the complement	Injection depth and input-dependent steering
κ	amplitude consistency	$1/(1 + \text{CV}(a)^2)$	Consistency of the shared teacher-induced shift

Theorem D.2 gives $\rho \geq \kappa/((1 + \eta)(1 + \varepsilon))$. Small principal intrusion, small complement residual, and consistent shared amplitudes are jointly sufficient for high recovery in the quadratic model. The experiments motivate these geometric quantities; their formal definitions and metric relationships are specified below.

PART I. SINGLE-VECTOR STEERING: CONDITIONS FOR HIGH RECOVERY

We introduce the local setting in Section D.1, derive the weighted least-squares formulation in Section D.2, and identify sufficient conditions for shared teacher structure in Section D.3. Section D.4 then quantifies recovery by a context-invariant vector.

D.1 SETUP AND NOTATION

We consider a frozen pretrained language model with decoder layers $\{f_\ell\}_{\ell=1}^L$, hidden width d , vocabulary size V , and unembedding matrix $U \in \mathbb{R}^{V \times d}$, which may be tied or untied. A context $x \sim \mathcal{D}$ consists of a token position and its prefix under the distillation distribution $\mathcal{D}$. The base model has hidden states and logits

$$\mathbf{h}_\ell(x) \in \mathbb{R}^d, \quad \mathbf{z}_0(x) = U\mathbf{h}_L(x) \in \mathbb{R}^V,$$

with $p_0(\cdot | x) = \text{softmax}(\mathbf{z}_0(x))$. Here $\mathbf{h}_L(x)$ is the representation supplied to the unembedding, including any final normalization. For an intervention before normalization, the downstream Jacobian defined below includes that operation. The teacher induces logits $\mathbf{z}_T(x)$ and distribution $p_T(\cdot | x)$. Vector steering adds a trainable, context-invariant offset $\delta_\ell \in \mathbb{R}^d$ at each controlled layer $\ell \in \mathcal{S} \subseteq \{1, \dots, L\}$: $\mathbf{h}_\ell(x) \mapsto \mathbf{h}_\ell(x) + \delta_\ell$, with all backbone weights frozen. We first analyze a single injection site and extend the notation to multiple sites in Remark D.2.

For a positive-semidefinite matrix A , write $\|v\|_A^2 = v^\top A v$ and $\langle u, v \rangle_A = u^\top A v$. The notation $\|\cdot\|_2$ denotes the Euclidean vector norm or matrix spectral norm, and $\lambda_i(\cdot)$ and $\sigma_i(\cdot)$ denote eigenvalues and singular values in descending order. All logarithms are natural and all vectors are columns. We analyze the expected forward-KL objective

$$\mathcal{L}(\delta) = \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(p_T(\cdot | x) \| p_\delta(\cdot | x)). \quad (32)$$

The context distribution is fixed for each local fitting problem; changing on-policy trajectories changes that problem. All expected quadratic energies are assumed finite.

Let $\mathcal{E} \subseteq \mathbb{R}^d$ be a fixed working subspace with orthonormal basis $T \in \mathbb{R}^{d \times h}$. Steering vectors and target corrections lie in $\mathcal{E}$, and inverses are taken in these coordinates, where the compressed metrics are positive definite. Setting $T = I_d$ recovers the full-space formulas. For singular metrics, the same construction is applied on a common identifiable subspace, with a consistent identification of targets across contexts.

Context-dependent and context-invariant quantities. We distinguish quantities that vary across contexts from the fixed objects being learned:

- **Context dependence** is indicated by (x) , as in $\mathbf{h}_\ell(x)$, $\mathbf{z}_T(x)$, $\mathbf{t}(x)$, $\Delta(x)$, $a(x)$, and $\xi(x)$, or by a subscript in J_x , F_x , G_x , and B_x .
- **Context-invariant objects** include trainable vectors δ_ℓ and $\mathbf{v}_\ell^{(k)}$, shared directions $\mathbf{s}$ and $\mathbf{w}$, averaged metric G , mean shift μ , projectors P, Q , and the constants used in the bounds. A trainable vector may change between optimization steps while remaining shared across contexts at each step.
- **Values and maps** are distinguished explicitly: context-indexed $\mathbf{h}$, $\mathbf{z}$, and Δ denote values, while maps such as $\phi: \mathbb{R} \rightarrow \mathbb{R}$ are introduced as functions.
- **Sequence and training notation** is introduced separately. Section D.3 distinguishes prompts q and complete responses Y from next-token contexts $x = (q, y_{<t})$; Part III defines batched activations and parameter-update indices.

The central fitting problem is therefore to approximate the context-dependent target cloud $\{\Delta(x)\}$ with one context-invariant vector in the relevant metric.

Assumptions. The local reduction uses Assumption D.1, and the constant-metric identities additionally use the exact form of Assumption D.3. Assumption D.2 specifies shared teacher structure, while Assumptions D.4–D.5 supply the geometric conditions for the three-constant bound. Part III’s local gradient-projection result requires only its own stated projection assumptions.

Assumption D.1 (Local / trust-region regime). *Steering operates in a neighborhood where the first-order logit expansion and second-order Fisher approximation to KL are accurate. Write*

$$J_x := \frac{\partial \mathbf{z}(x)}{\partial \mathbf{h}_\ell(x)} \in \mathbb{R}^{V \times d}, \quad \mathbf{z}_\delta(x) = \mathbf{z}_0(x) + J_x \delta + \mathbf{e}_z(x, \delta),$$

where J_x includes all downstream computation and $\mathbf{e}_z(x, \delta) = o(\|\delta\|_2)$. For $\mathbf{d}_\delta(x) = J_x \delta - \mathbf{t}(x)$, the contribution of $\mathbf{e}_z(x, \delta)$ to the quadratic KL approximation and the KL Taylor remainder are assumed to be $o(\|\mathbf{d}_\delta(x)\|_2^2)$. A sufficient local condition is $\mathbf{d}_\delta(x) \rightarrow 0$ and $\|\mathbf{e}_z(x, \delta)\|_2 = o(\|\mathbf{d}_\delta(x)\|_2)$, together with uniform bounds permitting interchange of approximation and expectation.

Assumption D.2 (Shared logit component). *The teacher logit shift $\mathbf{t}(x) := \mathbf{z}_T(x) - \mathbf{z}_0(x) \in \mathbb{R}^V$ admits the decomposition*

$$\mathbf{t}(x) = \underbrace{a(x)\mathbf{s}}_{\text{shared direction}} + \mathbf{r}_\perp(x), \quad (33)$$

where $\mathbf{s} \in \mathbb{R}^V$ is fixed, $\|\mathbf{s}\|_2 = 1$, $\mathbf{s}^\top \mathbf{s} = 0$, and $a(x) \geq 0$. The residual is Fisher-orthogonal to the shared direction at every context: $\langle \mathbf{s}, \mathbf{r}_\perp(x) \rangle_{F_x} = 0$. Its expected Fisher energy satisfies

$$\mathbb{E}_x \|\mathbf{r}_\perp(x)\|_{F_x}^2 \leq \eta_{\text{logit}} \mathbb{E}_x \|a(x)\mathbf{s}\|_{F_x}^2, \quad \eta_{\text{logit}} \ll 1. \quad (34)$$

The centered choice of $\mathbf{s}$ is justified in Remark D.3. Section D.3 derives sufficient conditions for this next-token structure from continuation-success profiles.

Assumption D.3 (Metric homogeneity). *The representational metric G_x from Definition D.1 is approximately context-independent. Writing $\bar{G} := \mathbb{E}_{x \sim \mathcal{D}}[G_x]$, we assume $G_x \approx \bar{G}$ and abbreviate $G := \bar{G}$. The matrix G is positive definite on the common working subspace, including the target shifts, admissible corrections, and projected components used below. Exact homogeneous-model identities use $G_x = G$; Remark D.1 treats perturbations. Consistency of the pulled-back shared direction $B_x \mathbf{s}$ is specified separately by ζ_B in Lemma D.5.*

Table 4: Recurring symbols for the distillation analysis. Context dependence is indicated by (x) or a subscript x ; batched training quantities are introduced in Part III.

Symbol	Type	Meaning
$x \sim \mathcal{D}$	context	Token position and prefix, $x = (q, y_{<t})$
q, Y	sequence level	Prompt and complete response, including termination; Section D.3
$\mathbf{h}_\ell(x) \in \mathbb{R}^d$	ctx.-dep.	Layer- ℓ hidden state
$\mathbf{z}_0(x), \mathbf{z}_T(x) \in \mathbb{R}^V$	ctx.-dep.	Base and teacher logits
$\mathbf{t}(x) = \mathbf{z}_T(x) - \mathbf{z}_0(x) \in \mathbb{R}^V$	ctx.-dep.	Teacher logit shift modulo ; Remark D.3
$F_x \in \mathbb{R}^{V \times V}$	ctx.-dep.	Fisher information at teacher logits; Equation (38)
$J_x \in \mathbb{R}^{V \times d}$	ctx.-dep.	Downstream logit Jacobian at the injection site
$G_x = J_x^\top F_x J_x \in \mathbb{R}^{d \times d}$	ctx.-dep.	Pullback metric; Definition D.1
$B_x = T(T^\top G_x T)^{-1} T^\top J_x^\top F_x$	ctx.-dep.	Pullback on the working subspace
$\Delta(x) \in \mathbb{R}^d$	ctx.-dep.	Target representation shift; Definition D.1
$a(x) \in \mathbb{R}_{\geq 0}$	ctx.-dep.	Amplitude in the stated shared-component decomposition
$\xi(x) \in \mathbb{R}^d$	ctx.-dep.	Residual within the complement
$\nu_0(x) \in [0, 1]$	ctx.-dep.	Conditional success probability; Equation (42)
$Z_\beta(x)$	ctx.-dep.	Conditional tilt normalizer; Equation (42)
$b(x), a_0(x), \mathbf{e}(x)$	ctx.-dep.	Baseline, modulation, and residual of the success profile
$\delta_\ell \in \mathbb{R}^d$	invariant	Trainable steering vector
$\mathbf{v}_\ell^{(k)} \in \mathbb{R}^d$	invariant	k -th sequentially trained orthogonal vector
$\mathbf{s} \in \mathbb{R}^V, \mathbf{w} \in \mathbb{R}^d$	invariant	Shared logit and representation directions
$G = \mathbb{E}_x[G_x]$	invariant	Averaged metric; Assumption D.3
$\mu = \mathbb{E}_x[\Delta(x)]$	invariant	Mean target shift; optimal vector when $G_x = G$
$C, P, Q = \mathbf{I} - P$	invariant	Centered activation covariance and its spectral projectors
$= \mathbf{I} - \top / V$	invariant	Vocabulary-centering projector, distinct from P
$c_\beta = e^{1/\beta} - 1$	invariant	Tilt strength for binary rewards
$\varepsilon, \eta, \kappa, \rho$	invariant	Intrusion bound, residual energy, amplitude consistency, and recovery
$\eta_{\text{logit}}, \theta, \zeta_B, \zeta_R$	invariant	Logit-residual and pullback-consistency constants

Assumption D.4 (Off-principal localization). *Let P be the orthogonal projector onto the top- m principal subspace of the activation covariance C from Definition D.3, and let $Q = \mathbf{I} - P$. For $\Delta(x) = P\Delta(x) + Q\Delta(x)$, assume an approximately G -orthogonal energy split,*

$$\mathbb{E}_x \|\Delta(x)\|_G^2 \approx \mathbb{E}_x \|P\Delta(x)\|_G^2 + \mathbb{E}_x \|Q\Delta(x)\|_G^2, \quad (35)$$

and the principal-to-complement energy bound

$$\mathbb{E}_x \|P\Delta(x)\|_G^2 \leq \varepsilon \mathbb{E}_x \|Q\Delta(x)\|_G^2, \quad 0 \leq \varepsilon \ll 1.$$

An exact G -orthogonal decomposition for every vector in the working subspace is equivalent to $P^\top GQ = 0$ there. This operator condition specifies compatibility between covariance and sensitivity; equality of averaged energies for a particular target distribution is a weaker statement. Remark D.1 treats approximate compatibility.

Assumption D.5 (Shared direction in the complement). *There is a fixed $\mathbf{w} \in \text{range}(Q)$ with $Q\mathbf{w} = \mathbf{w}$ and $\|\mathbf{w}\|_G = 1$ such that*

$$Q\Delta(x) = \underbrace{a(x)\mathbf{w}}_{\text{shared complement mode}} + \xi(x), \quad a(x) \geq 0, \quad (36)$$

for every context, with

$$\mathbb{E}_x[\xi(x)] = \mathbf{0}, \quad \mathbb{E}_x[a(x)\langle \mathbf{w}, \xi(x) \rangle_G] = 0. \quad (37)$$

The residual lies in the complement. Assume $0 < \mathbb{E}_x[a(x)^2] < \infty$ and finite target energy, and define

$$\kappa := \frac{(\mathbb{E}_x[a(x)])^2}{\mathbb{E}_x[a(x)^2]} \in (0, 1], \quad \eta := \frac{\mathbb{E}_x \|\xi(x)\|_G^2}{\mathbb{E}_x[a(x)^2]} \geq 0.$$

The amplitude-consistency factor satisfies $\kappa = 1$ if and only if $a(x)$ is constant almost surely, and $\kappa = 1/(1 + \text{CV}(a)^2)$, where $\text{CV}(a)^2 = \text{Var}_x(a(x))/(\mathbb{E}_x[a(x)])^2$. Nonnegativity fixes the sign of the shared amplitude; η measures residual energy relative to shared energy rather than rank.

The weighted orthogonality condition in Equation (37) removes the mixed term in Equation (59). Pointwise orthogonality, $\langle \mathbf{w}, \xi(x) \rangle_G = 0$, is a sufficient stronger condition. Here $a(x)$ refers to the representation-space

decomposition. It is a fixed positive rescaling of the logit-space amplitude in Assumption D.2 when the shared pullback is a context-independent nonzero vector in $\text{range}(Q)$ and the induced residual satisfies Equation (37). Otherwise the two amplitudes are defined separately, as discussed in Lemma D.5 and Remark D.5.

These assumptions specify three distinct ingredients: a local approximation regime, shared teacher-induced structure, and compatibility between the sensitivity metric and the activation geometry. Their roles in each result are stated explicitly.

Remark D.1 (Perturbations of the homogeneous model). (i) Varying metrics. *Without exact homogeneity, let $\bar{G}_\varepsilon = T^\top \bar{G} T$ and $\mathbf{b}_\varepsilon = T^\top \mathbb{E}_x[G_x \Delta(x)]$. The quadratic optimum is $\delta^* = T \bar{G}_\varepsilon^{-1} \mathbf{b}_\varepsilon$, with explained energy $S = \mathbf{b}_\varepsilon^\top \bar{G}_\varepsilon^{-1} \mathbf{b}_\varepsilon$ and residual energy $E_x - S$, where $E_x = \mathbb{E}_x \|\Delta(x)\|_{G_x}^2 > 0$. Thus $\rho_x = S/E_x$ has a generalized signal-to-residual interpretation. For a fixed target family, if $(1-\zeta)G \preceq G_x \preceq (1+\zeta)G$ uniformly with $0 \leq \zeta < 1$, the optimal varying- and constant-metric recovery ratios satisfy*

$$\frac{1-\zeta}{1+\zeta}(1-\rho_G) \leq 1-\rho_x \leq \frac{1+\zeta}{1-\zeta}(1-\rho_G).$$

This follows by bounding both the residual quadratic and the total target energy. Changes to the target family require a corresponding target-perturbation bound.

(ii) Approximately orthogonal splits. Suppose $|\langle u, v \rangle_G| \leq \gamma \|u\|_G \|v\|_G$ for $u \in \text{range}(P)$ and $v \in \text{range}(Q)$, with $0 \leq \gamma < 1$. Then

$$(1-\gamma)(\|Pz\|_G^2 + \|Qz\|_G^2) \leq \|z\|_G^2 \leq (1+\gamma)(\|Pz\|_G^2 + \|Qz\|_G^2).$$

Under the remaining hypotheses of Theorem D.2, its proof gives $\rho \geq \frac{1-\gamma}{1+\gamma} \frac{\kappa}{(1+\eta)(1+\varepsilon)}$. The correction accounts for both numerator and denominator.

(iii) Shared components and metric comparison. Corollary D.1 bounds logit residual energy, and Lemma D.5 bounds representation residual energy relative to a specified shared direction. Applying these estimates to η additionally requires the centering, weighted orthogonality, and localization conditions of Assumption D.5, or explicit bounds on their deviations. Part III relates Euclidean PC-energy measurements to sensitivity-weighted intrusion through Corollary D.4.

D.2 DISTILLATION REDUCES TO WEIGHTED LEAST SQUARES

The categorical distribution is an exponential family with natural parameter $\mathbf{z} \in \mathbb{R}^V$, log-partition function $A(\mathbf{z}) = \log \sum_{v=1}^V e^{z_v}$, and mean map $\nabla A(\mathbf{z}) = \text{softmax}(\mathbf{z})$. Writing $p = \text{softmax}(\mathbf{z})$, its Hessian is the Fisher information:

$$F(\mathbf{z}) = \nabla^2 A(\mathbf{z}) = \text{diag}(p) - pp^\top \succeq 0. \quad (38)$$

Throughout, $F_x := F(\mathbf{z}_T(x))$ is evaluated at the teacher logits. In particular, $F_x = \mathbf{0}$, which underlies the gauge invariance in Remark D.3.

Lemma D.1 (KL as a Bregman divergence and its local quadratic form). *For $\mathbf{z}, \mathbf{z}' \in \mathbb{R}^V$, $\text{KL}(\text{softmax}(\mathbf{z}') \parallel \text{softmax}(\mathbf{z})) = D_A(\mathbf{z} \parallel \mathbf{z}')$, where D_A is the Bregman divergence generated by A . Under Assumption D.1, with $\mathbf{t}(x) = \mathbf{z}_T(x) - \mathbf{z}_0(x)$ and $F_x = F(\mathbf{z}_T(x))$,*

$$\text{KL}(p_T(\cdot \mid x) \parallel p_\delta(\cdot \mid x)) = \frac{1}{2} (J_x \delta - \mathbf{t}(x))^\top F_x (J_x \delta - \mathbf{t}(x)) + o(\|J_x \delta - \mathbf{t}(x)\|_2^2). \quad (39)$$

Proof. Substituting A and $\nabla A = p$ into the Bregman divergence gives the KL identity. Taylor expansion in the first argument at $\mathbf{z}_T(x)$ yields the Fisher quadratic in $\mathbf{z}_\delta(x) - \mathbf{z}_T(x)$. The logit expansion and combined remainder condition in Assumption D.1 then give Equation (39). $\square$

Definition D.1 (Representational metric and target shift). *For each context, define the pullback metric, pullback operator, and target representation shift by*

$$G_x := J_x^\top F_x J_x \in \mathbb{R}^{d \times d}, \quad B_x := T(T^\top G_x T)^{-1} T^\top J_x^\top F_x \in \mathbb{R}^{d \times V},$$

$$\Delta(x) := \arg \min_{\mathbf{u} \in \mathcal{E}} \|J_x \mathbf{u} - \mathbf{t}(x)\|_{F_x}^2 = B_x \mathbf{t}(x),$$

where $T^\top G_x T$ is positive definite. For $T = I_d$, $B_x = G_x^{-1} J_x^\top F_x$. The metric G_x measures the output-distribution sensitivity of a hidden-state perturbation, while $\Delta(x)$ is the locally best hidden-state correction for reproducing the teacher's logit shift. The target varies with context; the trainable vector δ does not.

Proposition D.1 (Distillation as local weighted least squares). *Under Assumption D.1, the local quadratic surrogate is*

$$\mathcal{L}_{\text{quad}}(\delta) = \frac{1}{2} \mathbb{E}_{x \sim \mathcal{D}} (\delta - \Delta(x))^\top G_x (\delta - \Delta(x)) + C_{\text{irr}}, \quad (40)$$

where $C_{\text{irr}} \geq 0$ is independent of δ and represents teacher-shift energy outside the locally reachable logit space. The original KL objective differs from this surrogate by the remainder in Lemma D.1.

Proof. The quadratic term in Equation (39) is $\frac{1}{2} \|J_x \delta - \mathbf{t}(x)\|_{F_x}^2$. Write $\mathbf{t}(x) = J_x \Delta(x) + \mathbf{r}_x$. The least-squares normal equations give $T^\top J_x^\top F_x \mathbf{r}_x = 0$. Since $\delta - \Delta(x) \in \mathcal{E}$, the cross term vanishes and

$$\|J_x \delta - \mathbf{t}(x)\|_{F_x}^2 = \|J_x(\delta - \Delta(x))\|_{F_x}^2 + \mathbf{r}_x^\top F_x \mathbf{r}_x.$$

The first term is $\|\delta - \Delta(x)\|_{G_x}^2$. Taking expectations proves Equation (40), with $C_{\text{irr}} = \frac{1}{2} \mathbb{E}_x \|\mathbf{r}_x\|_{F_x}^2$. $\square$

Remark D.2 (Multiple injection sites). *For multiple sites, stack the steering vectors and let J_x be the joint Jacobian from all injection sites to the logits. Equation (40) then holds with $G_x = J_x^\top F_x J_x$. Direct injection at the unembedding input gives $J_x = U$; injection before normalization or other downstream operations includes their Jacobians.*

Proposition D.1 expresses steering as fitting a constant vector to $\{\Delta(x)\}$ in an output-sensitivity metric. Both the teacher shift and the frozen student’s downstream mapping determine this geometry. The recovery analysis takes the teacher displacement as given; the idealized optimum considered next supplies a separate account of how shared shift structure may arise.

D.3 THE RLVR TEACHER SHIFT: EXPONENTIAL TILTING AND SHARED STRUCTURE

We first derive the complete-response distribution of the KL-regularized RL optimum, then connect it to next-token shifts and their representation pullbacks. This separates three objects: sequence-level rewards, complete-response probability ratios, and token-level logit corrections. Shared next-token structure is characterized through continuation-success profiles.

Sequence notation and optimization setting. Write q for a prompt, Y for a complete response including termination, and $x = (q, y_{<t})$ for a next-token context. Responses terminate almost surely under the base policy. We consider bounded rewards, a positive KL coefficient β , and policies supported on the base policy’s support. The symbol π^* denotes the optimum of the stated regularized objective; empirical teacher checkpoints enter the local recovery analysis through their observed shifts $\mathbf{t}(x)$.

Lemma D.2 (KL-regularized RL optimum as an exponential tilt). *For a fixed prompt q , the objective*

$$\max_{\pi} \mathbb{E}_{Y \sim \pi(\cdot | q)} [r(q, Y)] - \beta \text{KL}(\pi(\cdot | q) \| p_0(\cdot | q)), \quad \beta > 0,$$

is strictly concave on its finite-KL domain and has the unique maximizer

$$\pi^*(Y | q) = \frac{p_0(Y | q) e^{r(q, Y)/\beta}}{Z_\beta(q)}, \quad Z_\beta(q) = \mathbb{E}_{Y \sim p_0(\cdot | q)} [e^{r(q, Y)/\beta}]. \quad (41)$$

Equivalently, $\log(\pi^(Y | q)/p_0(Y | q)) = r(q, Y)/\beta - \log Z_\beta(q)$.*

Proof. Bounded rewards make $Z_\beta(q)$ finite and positive. Substitution rewrites the objective as $\beta \log Z_\beta(q) - \beta \text{KL}(\pi(\cdot | q) \| \pi^*(\cdot | q))$. Nonnegativity of KL gives the unique maximum. Strict concavity follows from strict convexity of KL and linearity of the reward term. $\square$

The next result connects the complete-response identity in Equation (41) to the token-level shifts used in Definition D.1.

Lemma D.3 (From complete responses to next-token probabilities). *For $x = (q, y_{<t})$ with $p_0(y_{<t} | q) > 0$, define*

$$Z_\beta(x) := \mathbb{E}_{Y \sim p_0(\cdot | x)} [e^{r(q, Y)/\beta}], \quad \nu_0(x) := \Pr_{Y \sim p_0(\cdot | x)} (r(q, Y) = 1), \quad (42)$$

where the expectation and probability concern complete responses extending the prefix. For a supported next token v , write xv for the extended context. Then

$$\pi^*(v | x) = p_0(v | x) \frac{Z_\beta(xv)}{Z_\beta(x)}. \quad (43)$$

For binary rewards $r \in \{0, 1\}$ and $c_\beta := e^{1/\beta} - 1 > 0$,

$$\log \frac{\pi^*(v | x)}{p_0(v | x)} = \log[1 + c_\beta \nu_0(xv)] - \log[1 + c_\beta \nu_0(x)]. \quad (44)$$

At termination, the conditional quantities are evaluated using the resulting complete response.

Proof. Summing Equation (41) over complete responses extending the prefix gives

$$\pi^*(y_{<t} \mid q) = p_0(y_{<t} \mid q) \frac{Z_\beta(x)}{Z_\beta(q)}.$$

Taking the ratio of this identity for $y_{<t}v$ and $y_{<t}$ gives Equation (43). For binary rewards, $Z_\beta(x) = (1 - \nu_0(x)) + e^{1/\beta} \nu_0(x) = 1 + c_\beta \nu_0(x)$, which yields Equation (44). $\square$

Equation (44) identifies differences between candidate-token continuation-success probabilities as the source of relative logit changes. Its second term is token-independent. The tower property gives $\nu_0(x) = \sum_v p_0(v \mid x) \nu_0(xv)$, and concavity gives $\log[1 + c_\beta \nu_0(x)] \geq \sum_v p_0(v \mid x) \log[1 + c_\beta \nu_0(xv)]$. A shared correction across contexts therefore depends on shared structure in these success profiles, in addition to reward binarity.

Remark D.3 (Softmax gauge and centered logits). Since $\text{softmax}(\mathbf{z} + c) = \text{softmax}(\mathbf{z})$, logits are defined modulo the all-ones direction. Choose the centered gauge $:= \mathbf{I} - \mathbf{1}^\top / V$. Because $F_x = \mathbf{0}$,

$$F_x = F_x, \quad \|\mathbf{u}\|_{F_x} = \|\mathbf{u}\|_{F_x}, \quad B_x \mathbf{u} = B_x \mathbf{u} \quad \text{for all } \mathbf{u} \in \mathbb{R}^V. \quad (45)$$

Centering preserves Fisher seminorms, Fisher inner products, and $\Delta(x)$. For the idealized teacher, we may write $\mathbf{t}(x) = \log(\pi^*(\cdot \mid x)/p_0(\cdot \mid x))$ and choose $\mathbf{1}^\top \mathbf{s} = 0$. The vocabulary-centering projector is distinct from the activation-principal projector P .

Lemma D.4 (A sufficient condition for a shared logit component). Let $\mathbf{s} \in \mathbb{R}^V$ be fixed with $\|\mathbf{s}\|_2 = 1$ and $\mathbf{1}^\top \mathbf{s} = 0$. Suppose the continuation-success profiles satisfy

$$\nu_0(xv) = b(x) + a_0(x)s_v + e_v(x), \quad b(x) \in [0, 1], \quad a_0(x) \geq 0, \quad \nu_0(xv) \in [0, 1], \quad (46)$$

where $b(x)$ is a token-independent baseline, $a_0(x)$ a scalar modulation, and $\mathbf{e}(x)$ the residual profile. Define the total deviation $\mathbf{d}(x) := a_0(x)\mathbf{s} + \mathbf{e}(x)$. The centered logit shift then satisfies

$$\log \frac{\pi^*(\cdot \mid x)}{p_0(\cdot \mid x)} = \hat{a}(x)\mathbf{s} + \hat{\mathbf{r}}(x), \quad \hat{a}(x) := \frac{c_\beta a_0(x)}{1 + c_\beta b(x)} \geq 0, \quad (47)$$

with coordinatewise logarithms and ratios, and residual bound

$$\|\hat{\mathbf{r}}(x)\|_2 \leq c_\beta \|\mathbf{e}(x)\|_2 + \frac{c_\beta^2}{2} \|\mathbf{d}(x)\|_4^2, \quad \|\mathbf{d}(x)\|_4^2 := \left(\sum_{v=1}^V |d_v(x)|^4 \right)^{1/2}. \quad (48)$$

Proof. For $f(z) = \log(1 + c_\beta z)$ on $[0, 1]$, $|f'(z)| \leq c_\beta$ and $|f''(z)| \leq c_\beta^2$. Both $b(x)$ and $b(x) + d_v(x)$ lie in this interval, so Taylor's theorem gives

$$f(b(x) + d_v(x)) = f(b(x)) + f'(b(x))d_v(x) + R_v(x), \quad |R_v(x)| \leq \frac{1}{2} c_\beta^2 |d_v(x)|^2.$$

Applying to Equation (44) removes the token-independent terms. Since $\mathbf{s} = \mathbf{s}$, the residual is

$$\hat{\mathbf{r}}(x) = (f'(b(x))\mathbf{e}(x) + R(x)).$$

Using $\|\mathbf{s}\|_2 = 1$, $|f'(b(x))| \leq c_\beta$, and $\|R(x)\|_2 \leq \frac{1}{2} c_\beta^2 \|\mathbf{d}(x)\|_4^2$ gives Equation (48). $\square$

The profile condition specifies the shared structure needed to turn binary rewards into aligned token-level corrections. High relative alignment additionally requires the residual to be small compared with $\hat{a}(x)$. Remark D.4 gives an alternative bound for small positive β .

Corollary D.1 (Fisher-orthogonal shared-component decomposition). Assume Lemma D.4, $\|\mathbf{s}\|_{F_x} > 0$, and a fixed $0 \leq \theta < 1$ such that

$$\|\hat{\mathbf{r}}(x)\|_{F_x} \leq \theta \hat{a}(x) \|\mathbf{s}\|_{F_x} \quad \text{at every context considered.} \quad (49)$$

For $\mathbf{t}(x) = \hat{a}(x)\mathbf{s} + \hat{\mathbf{r}}(x)$, define

$$a(x) := \frac{\langle \mathbf{s}, \mathbf{t}(x) \rangle_{F_x}}{\|\mathbf{s}\|_{F_x}^2}, \quad \mathbf{r}_\perp(x) := \mathbf{t}(x) - a(x)\mathbf{s}. \quad (50)$$

Then $a(x) \geq (1 - \theta)\hat{a}(x) \geq 0$, $\langle \mathbf{s}, \mathbf{r}_\perp(x) \rangle_{F_x} = 0$, and

$$\mathbb{E}_x \|\mathbf{r}_\perp(x)\|_{F_x}^2 \leq \frac{\theta^2}{(1 - \theta)^2} \mathbb{E}_x \|a(x)\mathbf{s}\|_{F_x}^2. \quad (51)$$

The decomposition satisfies the relative energy bound with $\eta_{\text{logit}} = \theta^2/(1 - \theta)^2$, meeting the small-residual condition of Assumption D.2 when θ is sufficiently small.

Proof. Equation (50) gives $a(x) = \hat{a}(x) + \langle \mathbf{s}, \hat{\mathbf{r}}(x) \rangle_{F_x} / \|\mathbf{s}\|_{F_x}^2$. Cauchy–Schwarz and Equation (49) bound the second term by $\theta \hat{a}(x)$ in absolute value, yielding the amplitude bound. By construction, $a(x)\mathbf{s}$ is the Fisher-orthogonal projection of $\mathbf{t}(x)$ onto $\text{span}\{\mathbf{s}\}$. Its residual therefore satisfies

$$\|\mathbf{r}_\perp(x)\|_{F_x} \leq \|\mathbf{t}(x) - \hat{a}(x)\mathbf{s}\|_{F_x} = \|\hat{\mathbf{r}}(x)\|_{F_x} \leq \theta \hat{a}(x)\|\mathbf{s}\|_{F_x} \leq \frac{\theta}{1-\theta} a(x)\|\mathbf{s}\|_{F_x}.$$

Squaring and averaging proves Equation (51). Gauge invariance follows from Remark D.3. $\square$

Remark D.4 (Small positive β and uniform residual bounds). *A multiplicative success profile yields a bound without diverging constants as $\beta \downarrow 0$. Suppose $\nu_0(xv) \geq \nu_{\min} > 0$, $c_\beta \nu_{\min} \geq 1/\varrho$ for $0 < \varrho \leq 1$, and*

$$\log \nu_0(xv) = \tilde{b}(x) + a_0(x)s_v + \tilde{e}_v(x), \quad a_0(x) \geq 0. \quad (52)$$

Using $\log(1 + c_\beta z) = \log c_\beta + \log z + \log(1 + 1/(c_\beta z))$ gives

$$\log \frac{\pi^*(\cdot | x)}{p_0(\cdot | x)} = a_0(x)\mathbf{s} + \tilde{\mathbf{e}}(x) + \boldsymbol{\epsilon}(x), \quad \|\boldsymbol{\epsilon}(x)\|_2 \leq \sqrt{V} \varrho.$$

The bound follows from $0 \leq \log(1 + 1/(c_\beta \nu_0(xv))) \leq \varrho$ and contraction of $\cdot$. Corollary D.1 applies when its relative Fisher-residual condition holds with amplitude $a_0(x)$ and residual $\tilde{\mathbf{e}}(x) + \boldsymbol{\epsilon}(x)$. This regime concerns an idealized optimum with a small positive KL-penalty coefficient; it is distinct from assuming a small realized KL divergence or imposing a tilt identity on a zero-penalty checkpoint.

The shared component has so far been specified in logit space. Its usefulness for steering also depends on how consistently the frozen student maps that component back to representation space.

Lemma D.5 (Consistency of the representation pullback). *Retain Definition D.1 and Assumption D.2, with $\mathbf{t}(x) = a(x)\mathbf{s} + \mathbf{r}_\perp(x)$, $B_x = T(T^\top G_x T)^{-1} T^\top J_x^\top F_x$, and $\boldsymbol{\Delta}(x) = B_x \mathbf{t}(x)$. For a fixed nonzero $\mathbf{w}_0 \in \mathcal{E}$, define $\mathbf{e}_h(x) := \boldsymbol{\Delta}(x) - a(x)\mathbf{w}_0$. Then*

$$\mathbf{e}_h(x) = a(x)(B_x \mathbf{s} - \mathbf{w}_0) + B_x \mathbf{r}_\perp(x). \quad (53)$$

Let G be a fixed positive definite metric, set $E_0 := \mathbb{E}_x[a(x)^2]\|\mathbf{w}_0\|_G^2 > 0$, and suppose

$$\mathbb{E}_x[a(x)^2\|B_x \mathbf{s} - \mathbf{w}_0\|_G^2] \leq \zeta_B E_0, \quad \mathbb{E}_x\|B_x \mathbf{r}_\perp(x)\|_G^2 \leq \zeta_R E_0 \quad (54)$$

for nonnegative ζ_B and ζ_R . Then

$$\mathbb{E}_x\|\mathbf{e}_h(x)\|_G^2 \leq 2(\zeta_B + \zeta_R)E_0. \quad (55)$$

The two constants quantify variation of the shared pullback and energy of the pulled-back residual, respectively.

Proof. Substituting the logit decomposition into $\boldsymbol{\Delta}(x) = B_x \mathbf{t}(x)$ and adding and subtracting $a(x)\mathbf{w}_0$ gives Equation (53). Apply $\|\mathbf{u} + \mathbf{v}\|_G^2 \leq 2\|\mathbf{u}\|_G^2 + 2\|\mathbf{v}\|_G^2$, take expectations, and use Equation (54). $\square$

Under exact homogeneity, logit residual energy also bounds the pulled-back residual. Let $A_x = F_x^{1/2} J_x T$ and $\mathcal{P}_x = A_x(A_x^\top A_x)^{-1} A_x^\top$. For $G_x = G$,

$$\|B_x \mathbf{r}_\perp(x)\|_G^2 = (F_x^{1/2} \mathbf{r}_\perp(x))^\top \mathcal{P}_x (F_x^{1/2} \mathbf{r}_\perp(x)) \leq \|\mathbf{r}_\perp(x)\|_{F_x}^2. \quad (56)$$

The tight residual ratio therefore satisfies $\zeta_R \leq \eta_{\text{logit}} \mathbb{E}_x\|a(x)\mathbf{s}\|_{F_x}^2 / E_0$.

Remark D.5 (Shared-direction consistency under homogeneous metrics). *Equation (56) controls residual energy, while ζ_B controls variation of $B_x \mathbf{s} = T(T^\top G T)^{-1} T^\top J_x^\top F_x \mathbf{s}$. These are distinct requirements even when $G_x = G$. For a consistent shared direction $\mathbf{w}_0$, the normalization $\mathbf{w} = \mathbf{w}_0 / \|\mathbf{w}_0\|_G$ and $a_h(x) = a(x)\|\mathbf{w}_0\|_G$ gives a normalized representation component. Applying Assumption D.5 additionally requires $Q\mathbf{w} = \mathbf{w}$, zero residual mean, and the stated weighted orthogonality.*

From reward structure to representation recovery. The preceding results establish a sequence of sufficient conditions: exponential reward tilting yields the exact next-token relation; shared success profiles yield a shared logit component; and a consistent pullback yields a shared representation component. The final recovery bound additionally uses off-principal localization and the centered complement decomposition in Assumptions D.4–D.5. This separates the source of teacher structure from the geometric requirements on the student and intervention.

Connection to practical RL teachers. The exponential-tilt analysis concerns the exact optimum with $\beta > 0$. For trained checkpoints, including DAPO checkpoints with zero explicit KL coefficient, the local least-squares and recovery results instead take $\mathbf{t}(x)$ as given and apply under their stated approximation and structure assumptions. Small-step training and clipping motivate examining a local regime; the teacher need not be identified with an exact regularized optimum.

D.4 SINGLE-VECTOR RECOVERY: AN IDENTITY AND A LOWER BOUND

We now quantify the best constant-vector fit. The general quadratic optimum is metric-weighted. Under exact homogeneity, its explained target energy has the signal-to-fluctuation form in Section D.4.1 and the geometric lower bound in Section D.4.2.

Proposition D.2 (Optimal steering vector). *Assume finite target energy and finite C_{irr} . If $T^\top \bar{G} T$ is positive definite, where $\bar{G} = \mathbb{E}_x[G_x]$, the unique minimizer over $\mathcal{E}$ is $\delta^* = T(T^\top \bar{G} T)^{-1} T^\top \mathbb{E}_x[G_x \Delta(x)]$. For $T = I_d$, this becomes $\bar{G}^{-1} \mathbb{E}_x[G_x \Delta(x)]$. Under exact homogeneity $G_x = G$,*

$$\delta^* = \mu := \mathbb{E}_{x \sim \mathcal{D}}[\Delta(x)].$$

Under approximate homogeneity, the ordinary mean is an approximation to the metric-weighted optimum.

Proof. Writing $\delta = Tz$, the normal equations are $T^\top \bar{G} T z = T^\top \mathbb{E}_x[G_x \Delta(x)]$. Positive definiteness gives the unique solution. When $G_x = G$, the solution is $\delta = \mathbb{E}_x[\Delta(x)] \in \mathcal{E}$. $\square$

Definition D.2 (Recovery ratio). *In the constant-metric quadratic model, assume $0 < \mathbb{E}_x \|\Delta(x)\|_G^2 < \infty$ and define*

$$\rho := \frac{\mathbb{E}_x \|\Delta(x)\|_G^2 - \mathbb{E}_x \|\Delta(x) - \delta^*\|_G^2}{\mathbb{E}_x \|\Delta(x)\|_G^2} \in [0, 1].$$

This is the fraction of locally representable target energy explained by the optimal constant vector, excluding C_{irr} . Thus $\rho = 1$ represents an exact fit to the target shifts in the surrogate. Exact teacher reproduction additionally requires local logit reachability and negligible approximation error. This energy ratio is distinct from empirical task-score gain recovery; Remark D.1(i) defines its varying-metric counterpart.

D.4.1 SIGNAL-TO-FLUCTUATION FORM

Theorem D.1 (Recovery as a signal-to-fluctuation ratio). *In the quadratic model with $G_x = G$, let $\tilde{\Delta}(x) = \Delta(x) - \mu$, so $\mathbb{E}_x[\tilde{\Delta}(x)] = \mathbf{0}$. Then*

$$\boxed{\begin{aligned} \rho &= \frac{\|\mu\|_G^2}{\|\mu\|_G^2 + \mathbb{E}_x \|\tilde{\Delta}(x)\|_G^2} = \frac{\text{SNR}}{1 + \text{SNR}}, \\ \text{SNR} &:= \frac{\|\mu\|_G^2}{\mathbb{E}_x \|\tilde{\Delta}(x)\|_G^2}. \end{aligned}} \quad (57)$$

If the SNR denominator vanishes, use $\text{SNR} = +\infty$ and $\rho = 1$. For this energy ratio, $\rho \geq 0.85$ if and only if $\text{SNR} \geq 0.85/(1 - 0.85) \approx 5.7$.

Proof. Proposition D.2 gives $\delta^* = \mu$. Since the fluctuation has zero mean, $\mathbb{E}_x \|\Delta(x)\|_G^2 = \|\mu\|_G^2 + \mathbb{E}_x \|\tilde{\Delta}(x)\|_G^2$. Substitution into Definition D.2 gives Equation (57). $\square$

Corollary D.2 (Spectral upper bound). *In the same model, let $\mathbf{u}(x) = G^{1/2} \Delta(x)$, $\mathbf{m} = \mathbb{E}_x[\mathbf{u}(x)] = G^{1/2} \mu$, and $\Sigma = \mathbb{E}_x[\mathbf{u}(x) \mathbf{u}(x)^\top]$. For a weighted data operator M satisfying $\Sigma = M M^\top$,*

$$\rho = \frac{\|\mathbf{m}\|_2^2}{\text{tr } \Sigma} \leq \frac{\lambda_1(\Sigma)}{\text{tr } \Sigma} = \frac{\sigma_1^2(M)}{\sum_i \sigma_i^2(M)}.$$

For $\mathbf{m} \neq \mathbf{0}$, equality holds if and only if $\mathbf{m}$ lies in the leading eigenspace of Σ and $\mathbf{m}^\top \text{Cov}(\mathbf{u}(x)) \mathbf{m} = 0$. For $\mathbf{m} = \mathbf{0}$ and $\text{tr } \Sigma > 0$, the inequality is strict.

Proof. Since $\mathbf{m} \mathbf{m}^\top \preceq \Sigma$, $\|\mathbf{m}\|_2^2 \leq \lambda_1(\Sigma)$. For $v = \mathbf{m}/\|\mathbf{m}\|_2$,

$$\lambda_1(\Sigma) \geq v^\top \Sigma v = \|\mathbf{m}\|_2^2 + v^\top \text{Cov}(\mathbf{u}(x)) v \geq \|\mathbf{m}\|_2^2.$$

Equality requires equality at both steps, giving the stated conditions. The denominator is $\mathbb{E}_x \|\Delta(x)\|_G^2 = \text{tr } \Sigma$. $\square$

The leading spectral share bounds constant-vector recovery, but does not generally equal it. Collinear targets can have unit leading spectral share while strongly varying amplitudes reduce recovery by a fixed vector. Part II contrasts this fitting problem with shared-subspace approximation using context-dependent coefficients.

D.4.2 EXPLICIT THREE-CONSTANT BOUND

Assumptions D.4–D.5 express favorable recovery conditions through principal energy, complement residual energy, and shared-amplitude consistency.

Theorem D.2 (Off-principal single-direction bound). *In the quadratic model, assume $G_x = G$, $P^\top GQ = 0$, and the localization and shared-component conditions of Assumptions D.4–D.5. Then*

$$\rho \geq \frac{\kappa}{(1+\eta)(1+\varepsilon)}. \quad (58)$$

Energy decomposition. The weighted orthogonality condition in Equation (37) removes the mixed term in

$$\mathbb{E}_x \|a(x)\mathbf{w} + \boldsymbol{\xi}(x)\|_G^2 = \mathbb{E}_x [a(x)^2] \|\mathbf{w}\|_G^2 + \mathbb{E}_x \|\boldsymbol{\xi}(x)\|_G^2 + 2\mathbb{E}_x [a(x)\langle \mathbf{w}, \boldsymbol{\xi}(x) \rangle_G]. \quad (59)$$

Together with $P^\top GQ = 0$ and the zero-mean residual, this gives the exact decomposition used below. Remark D.1 provides the corresponding perturbation bounds.

Proof. Theorem D.1 gives $\rho = \|\boldsymbol{\mu}\|_G^2 / \mathbb{E}_x \|\boldsymbol{\Delta}(x)\|_G^2$. Since $\mathbb{E}_x [\boldsymbol{\xi}(x)] = \mathbf{0}$, $Q\boldsymbol{\mu} = \mathbb{E}_x [a(x)]\mathbf{w}$. The G -orthogonal split and $\|\mathbf{w}\|_G = 1$ imply

$$\|\boldsymbol{\mu}\|_G^2 \geq \|Q\boldsymbol{\mu}\|_G^2 = (\mathbb{E}_x [a(x)])^2 = \kappa \mathbb{E}_x [a(x)^2].$$

Meanwhile,

$$\mathbb{E}_x \|\boldsymbol{\Delta}(x)\|_G^2 \leq (1+\varepsilon) \mathbb{E}_x \|Q\boldsymbol{\Delta}(x)\|_G^2 = (1+\varepsilon)(1+\eta) \mathbb{E}_x [a(x)^2],$$

where Equations (37) and (59) give the last equality. Dividing proves Equation (58). $\square$

Corollary D.3 (Coefficient-of-variation form). *Under the same conditions,*

$$\rho \geq \frac{1}{(1 + \text{CV}(a)^2)(1+\eta)(1+\varepsilon)},$$

since $\kappa = 1/(1 + \text{CV}(a)^2)$.

Transition to multi-vector steering. The bound identifies a sufficient regime for high constant-vector recovery: low principal energy, a small complement residual, and consistent shared amplitudes. We next characterize the best fit after previously learned directions are excluded and distinguish this intervention protocol from adaptive subspace approximation.

PART II. MULTI-VECTOR STEERING: THE GEOMETRY OF THE RESIDUAL SUBSPACE

Part I characterized the best constant correction. Section D.5 extends this analysis to subspace-constrained fitting and relates it to sequential orthogonal steering. Section D.6 then connects activation principal structure, output sensitivity, and weight-induced activation changes.

D.5 SEQUENTIAL-ORTHOGONAL STEERING AND CONSTRAINED FITTING

The empirical procedure learns $\mathbf{v}^{(0)}, \dots, \mathbf{v}^{(K-1)}$ sequentially. At each stage, only the current multi-layer direction is injected; earlier directions define layer-wise orthogonality constraints, and reprojection enforces those constraints after each optimizer step. Each direction is evaluated independently. We analyze this protocol as constrained constant-vector fitting, separately from spectral approximation with adaptive coefficients.

Theorem D.3 (Constant-vector recovery under a subspace constraint). *In the constant-metric quadratic model, let $\mathcal{V}_k \subseteq \mathcal{E}$ be the feasible subspace at stage k , and let R_k have full-column-rank columns spanning it. For the single-site sequential protocol, $\mathcal{V}_k = \mathcal{E} \cap \text{span}\{\mathbf{v}^{(j)} : j < k\}^\perp$. The exact constrained minimizer is*

$$\mathbf{v}^{(k)} = R_k(R_k^\top G R_k)^{-1} R_k^\top G \boldsymbol{\mu}.$$

Writing $\rho^{(k)}$ for its standalone energy recovery and $E_{\text{tar}} = \mathbb{E}_x \|\boldsymbol{\Delta}(x)\|_G^2 > 0$,

$$\rho^{(k)} = \frac{\|\mathbf{v}^{(k)}\|_G^2}{E_{\text{tar}}}, \quad \mathcal{V}_{k+1} \subseteq \mathcal{V}_k \implies \rho^{(k+1)} \leq \rho^{(k)}. \quad (60)$$

For $G = cI$, $c > 0$, and $\boldsymbol{\mu} \neq 0$, exact first-stage optimization gives $\mathbf{v}^{(0)} = \boldsymbol{\mu}$ and exact second-stage optimization in its Euclidean orthogonal complement gives $\mathbf{v}^{(1)} = \mathbf{0}$.

Proof. The parallel-axis identity separates the objective into a constant fluctuation term and $\|\mathbf{v} - \boldsymbol{\mu}\|_G^2$. Writing $\mathbf{v} = R_k z$ gives the normal equation $R_k^\top G(R_k z - \boldsymbol{\mu}) = 0$ and hence the minimizer. Its residual is G -orthogonal to $\mathcal{V}_k$, so $\langle \mathbf{v}^{(k)}, \boldsymbol{\mu} \rangle_G = \|\mathbf{v}^{(k)}\|_G^2$. Expanding the loss improvement gives Equation (60), and nesting the feasible spaces cannot improve the optimal fit. When $G = cI$, the second space is orthogonal to $\boldsymbol{\mu}$, whose projection is zero. A zero-dimensional feasible space has the unique minimizer zero; a zero vector adds no further orthogonality constraint. $\square$

For multiple sites, the feasible space is the product of layer-wise feasible spaces intersected with the common working subspace. The theorem applies to the stacked vectors and the full metric G , including its cross-layer coupling blocks.

Adaptive subspace approximation. For comparison, a shared subspace with context-dependent coefficients admits a spectral characterization. Let $1 \leq K \leq \dim \mathcal{E}$ and $\mathbf{u}(x) = G^{1/2} \boldsymbol{\Delta}(x)$. Then

$$\min_{\substack{Q_K^\top Q_K = I_K \\ \text{range}(Q_K) \subseteq G^{1/2} \mathcal{E}}} \mathbb{E}_x \|\mathbf{u}(x) - Q_K Q_K^\top \mathbf{u}(x)\|_2^2 = \text{tr} \Sigma - \sum_{i=1}^K \lambda_i(\Sigma). \quad (61)$$

The captured energy is $\text{tr}(Q_K^\top \Sigma Q_K)$, which is maximized by the leading eigenspace. The fitted coefficients $Q_K^\top \mathbf{u}(x)$ vary with context, yielding explained-energy fraction $\sum_{i=1}^K \lambda_i(\Sigma) / \text{tr} \Sigma$. This is a different intervention class from independently evaluated constant vectors. Additional layer-wise structural constraints may further restrict its attainable fit.

Connection to the measured steering directions. The standalone effectiveness of later orthogonal directions demonstrates that the nonlinear steering procedure admits multiple useful interventions. A decline with extraction order is consistent with increasingly restricted feasible spaces, while the idealized isotropic quadratic model in Theorem D.3 gives zero recovery after the first exact direction. The empirical setting additionally includes metric anisotropy, evolving sampled contexts, nonlinear responses, and approximate optimization, and evaluates task scores rather than surrogate energy. Thus, the observed slower decline in larger models characterizes the persistence of independently effective interventions. Equation (61) distinguishes this observation from the spectral gains achievable with context-dependent coefficients.

D.6 EFFECTIVE DIRECTIONS AND THE PRINCIPAL SUBSPACE

The main-text directions place approximately 1%–2% of their Euclidean energy in the top-10% activation principal subspace, below the approximately 10% isotropic reference. To relate this localization to the recovery analysis, we distinguish activation variance from output sensitivity and derive a sensitivity-weighted energy bound.

Definition D.3 (Variance and sensitivity metrics). Let $\bar{\mathbf{h}}_\ell = \mathbb{E}_x[\mathbf{h}_\ell(x)]$ and define

$$C := \mathbb{E}_x[(\mathbf{h}_\ell(x) - \bar{\mathbf{h}}_\ell)(\mathbf{h}_\ell(x) - \bar{\mathbf{h}}_\ell)^\top], \quad G := \mathbb{E}_x[G_x].$$

The leading eigenspace of the centered covariance C defines P , with $Q = I - P$. The matrix G measures output sensitivity; at the unembedding input it is $U^\top \mathbb{E}_x[F_x]U$. An uncentered implementation instead uses $\mathbb{E}_x[\mathbf{h}_\ell(x)\mathbf{h}_\ell(x)^\top] = C + \bar{\mathbf{h}}_\ell \bar{\mathbf{h}}_\ell^\top$ and must report and use its corresponding basis consistently.

Proposition D.3 (Sensitivity-weighted principal-energy bound). Define

$$g_P = \sup_{v \in \text{range}(P) \setminus \{0\}} \frac{v^\top G v}{\|v\|_2^2}, \quad g_Q = \inf_{v \in \text{span}\{Q\boldsymbol{\Delta}(x)\} \setminus \{0\}} \frac{v^\top G v}{\|v\|_2^2} > 0.$$

If $\mathbb{E}_x \|Q\boldsymbol{\Delta}(x)\|_2^2 > 0$, the actual sensitivity-weighted intrusion satisfies

$$e_G := \frac{\mathbb{E}_x \|P\boldsymbol{\Delta}(x)\|_G^2}{\mathbb{E}_x \|Q\boldsymbol{\Delta}(x)\|_G^2} \leq \frac{g_P \mathbb{E}_x \|P\boldsymbol{\Delta}(x)\|_2^2}{g_Q \mathbb{E}_x \|Q\boldsymbol{\Delta}(x)\|_2^2}.$$

A small sensitivity ratio together with a controlled Euclidean energy split therefore gives a small admissible bound $\varepsilon \geq e_G$ in Assumption D.4.

Proof. For every context, $\|P\boldsymbol{\Delta}(x)\|_G^2 \leq g_P \|P\boldsymbol{\Delta}(x)\|_2^2$ and $\|Q\boldsymbol{\Delta}(x)\|_G^2 \geq g_Q \|Q\boldsymbol{\Delta}(x)\|_2^2$. Taking expectations and dividing proves the result. $\square$

Low-sensitivity principal directions. Softmax is invariant to constant logit shifts. If $J_x v = c(x) + e(x)$, then $v^\top G_x v = \|e(x)\|_{F_x}^2$, so a small Fisher residual makes v a low-sensitivity direction. At the unembedding input, $J_x = U$. A small g_P requires this behavior uniformly over the principal subspace. The condition concerns sensitivity, whereas measured PC fractions concern Euclidean localization; Corollary D.4 relates the two under compatible metrics. The shared direction’s localization, $Q\mathbf{w} = \mathbf{w}$, remains the geometric condition specified in Assumption D.5.

D.6.1 A WEIGHT-SPACE ACCOUNT OF ACTIVATION LEAKAGE

We complement the representation-space analysis by relating a layer’s weight displacement to its first-order activation change.

Weight-principal components. For $W \in \mathbb{R}^{d \times d'}$ with singular value decomposition $W = \sum_i s_i \mathbf{p}_i \mathbf{q}_i^\top$, define

$$P_W := \sum_{i \leq k} \mathbf{p}_i \mathbf{p}_i^\top.$$

The split $\Delta W = P_W \Delta W + (I - P_W) \Delta W$ separates principal weight components from their complement. Update magnitude and orientation play different roles: the perturbation bound $|s_i(W + \Delta W) - s_i(W)| \leq \|\Delta W\|_2$ controls absolute spectral changes, while the following lemma tracks how the two components contribute to activation leakage.

Lemma D.6 (Weight components and first-order activation leakage). *Consider $\mathbf{h}_\ell(x) = \phi(W \mathbf{c}(x))$, holding $\mathbf{c}(x)$ fixed while perturbing W . Let $D(x) = \text{diag}(\phi'(W \mathbf{c}(x)))$, with bounded derivative at the evaluated points. For the first-order shift $\Delta_{\mathbf{h}}(x) := D(x) \Delta W \mathbf{c}(x)$,*

$$\|P \Delta_{\mathbf{h}}(x)\|_2 \leq \underbrace{\|D(x)\|_2 \|P_W \Delta W\|_2 \|\mathbf{c}(x)\|_2}_{\text{on-principal weight mass}} + \underbrace{\|PD(x)(\mathbf{I} - P_W)\|_2 \|\Delta W\|_2 \|\mathbf{c}(x)\|_2}_{\text{cross-subspace leakage}}. \quad (62)$$

For nonzero total shift, if the right-hand side is at most $\xi \|\Delta_{\mathbf{h}}(x)\|_2$, then $\|P \Delta_{\mathbf{h}}(x)\|_2 / \|\Delta_{\mathbf{h}}(x)\|_2 \leq \xi$.

Proof. Splitting ΔW gives

$$P \Delta_{\mathbf{h}}(x) = PD(x) P_W \Delta W \mathbf{c}(x) + PD(x) (\mathbf{I} - P_W) \Delta W \mathbf{c}(x).$$

The triangle inequality, submultiplicativity, and $\|P\|_2 \leq 1$ prove Equation (62). Dividing by $\|\Delta_{\mathbf{h}}(x)\|_2 > 0$ yields the relative bound. $\square$

The leakage factor separates into basis mismatch and Jacobian mixing: $PD(x)(\mathbf{I} - P_W) = (P - P_W)D(x)(\mathbf{I} - P_W) + P_W D(x)(\mathbf{I} - P_W)$. The bound therefore depends on both the weight-activation alignment and the intervening transformation. Extending it to the nonlinear activation change introduces a Taylor remainder; updates to earlier layers additionally change $\mathbf{c}(x)$. This first-order weight-induced shift and the distillation target $\Delta(x)$ describe different objects. Part III directly analyzes the parameter-to-activation mapping relevant to Alpha-Stabler.

PART III. GEOMETRIC ANALYSIS OF ALPHA-STABLER

Parts I–II characterize when fixed activation interventions can reproduce RL-induced gains and examine the geometry of the resulting steering directions. We now analyze the parameter-training procedure motivated by these observations. Section D.7 relates token-level PSI to principal-subspace energy, establishes the exact properties of backward gradient projection, and derives a local relationship between projected gradients and activation changes. Section D.8 summarizes the theoretical and empirical results.

D.7 PRINCIPAL-SUBSPACE MONITORING AND GRADIENT CONTROL

Warm-up estimation and fixed reference geometry. During the first $T_{\text{warm}} = 50$ parameter updates, the actor follows ordinary RL without gradient projection. Actor-generated rollouts are evaluated by the frozen initial base model to estimate its token-level activation covariance. At the end of warm-up, let $\mathcal{U}_\ell \in \mathbb{R}^{d \times r}$ contain the leading orthonormal eigenvectors at layer ℓ , with $r = \lceil 0.10d \rceil$. This is the basis denoted by $U_{\ell,r}$ in the main text; the retained fraction refers to hidden dimensions rather than explained variance. Define $P_\ell = \mathcal{U}_\ell \mathcal{U}_\ell^\top$ and $Q_\ell = \mathbf{I} - P_\ell$. These bases remain fixed after warm-up.

The analysis conditions on the resulting reference projectors. During each controlled backward pass and optimizer step, both projectors and control flags are held fixed, with no gradients propagated through their estimation or selection.

Token-level principal-subspace intrusion. On identical token sequences, stack the current policy’s activation differences from the frozen base model as rows of $X_\ell^{(t)} \in \mathbb{R}^{N \times d}$. Consistent with Equation (??), the layer-wise statistic is

$$\text{PSI}_\ell^{(t)} = \frac{\|X_\ell^{(t)} \mathcal{U}_\ell\|_F^2}{\|X_\ell^{(t)}\|_F^2 + \epsilon_{\text{num}}}, \quad \epsilon_{\text{num}} > 0. \quad (63)$$

Here ϵ_{num} corresponds to the numerical stabilizer ϵ in the main-text PSI definition and is distinct from the theoretical intrusion bound ϵ . Projection energies are summed over tokens before normalization to avoid cancellation between differently oriented shifts. The aggregate PSI similarly sums principal and total shift energies across monitored layers before taking their ratio.

Corollary D.4 (Energy interpretation and metric comparison). *Fix a layer and suppress its indices. Let $A = \|XP\|_F^2$, $B = \|XQ\|_F^2$, and $A + B > 0$. The unstabilized energy fraction is $\psi_0 = A/(A + B)$. If $B > 0$, define $e_E = A/B$; then*

$$\psi_0 = \frac{e_E}{1 + e_E}, \quad 0 \leq \psi_0 - \text{PSI} = \psi_0 \frac{\epsilon_{\text{num}}}{A + B + \epsilon_{\text{num}}}. \quad (64)$$

For $B > 0$ and the same shifts, define the metric-weighted ratio $e_G = \sum_i \|P\Delta_i\|_G^2 / \sum_i \|Q\Delta_i\|_G^2$. Suppose $0 < m_G \leq M_G$ satisfy $m_G \|v\|_2^2 \leq \|v\|_G^2 \leq M_G \|v\|_2^2$ on the span of the projected shifts, and write $\chi = M_G/m_G$. Then

$$\frac{e_G}{\chi} \leq e_E \leq \chi e_G. \quad (65)$$

Consequently, if the same shift collection satisfies $e_G \leq \epsilon$, then $\text{PSI} \leq \psi_0 \leq \chi\epsilon/(1 + \chi\epsilon)$.

Proof. Orthogonality gives $\|X\|_F^2 = A + B$ and $\|XU\|_F^2 = A$. Substitution proves Equation (64). The metric bounds imply $m_G A \leq \sum_i \|P\Delta_i\|_G^2 \leq M_G A$ and the analogous inequalities for B . Taking ratios proves Equation (65); monotonicity of $u/(1 + u)$ gives the last claim. If $B = 0$ and $A > 0$, $\psi_0 = 1$; if $A + B = 0$, the ratio contains no directional information. $\square$

Corollary D.4 describes the actual activation shifts in X . Connecting these shifts to the distillation targets $\Delta(x)$ in Parts I–II requires compatible shift distributions and metric bounds. The parameter ϵ retains its definition as the ratio upper bound in Assumption D.4.

Predictor and threshold calibration. During warm-up, checks occur every $M = 3$ parameter updates. We retain the detached actor-to-base activation-shift matrix when $N > 0$ and $\|X_\ell^{(t)}\|_F^2/N > \eta_{\min}$, using the actor state at the original check. Once the final reference bases are estimated, we evaluate these retained shifts using Equation (63). Let $\mathcal{W}_\ell$ denote the resulting collection of valid warm-up PSI values, as in Algorithm 1. We compute

$$\begin{aligned} m_\ell &= \text{median}_{\psi \in \mathcal{W}_\ell} \psi, & s_\ell &= 1.4826 \text{ median}_{\psi \in \mathcal{W}_\ell} |\psi - m_\ell| + \epsilon_{\text{num}}, \\ \tau_\ell^{\text{on}} &= m_\ell + 3s_\ell, & \tau_\ell^{\text{safe}} &= m_\ell + 2s_\ell. \end{aligned} \quad (66)$$

Calibration uses an initially stable warm-up window and the same fixed bases as subsequent monitoring. Training stops for recalibration if any $\mathcal{W}_\ell$ is empty or $0 < \tau_\ell^{\text{safe}} < \tau_\ell^{\text{on}} < 1$ fails. After calibration, the bases and thresholds remain fixed and the retained warm-up records can be released.

Monitoring continues every M updates. Valid observations update the moving average as $\bar{\psi}_\ell \leftarrow 0.95\bar{\psi}_\ell + 0.05\text{PSI}_\ell$, initialized at m_ℓ . A flag $a_\ell \in \{0, 1\}$, initially zero, activates after $p = 3$ consecutive valid checks with smoothed PSI above the warning threshold and releases when smoothed PSI falls below the safe threshold. The consecutive-warning count resets on a valid check that does not exceed the warning threshold, or on an invalid check; invalid checks leave the moving average and flag unchanged. Flags are held between checks and fixed during each backward pass and optimizer step. Thus, the Predictor combines a shared reference geometry with smoothing, persistence, and hysteresis.

Controller: backward-only gradient projection. Let $G_\ell^{(t)} \in \mathbb{R}^{N \times d}$ denote the gradient arriving at a monitored layer output before its local control operation. This activation gradient is distinct from the sensitivity metric G in Parts I–II. With several controlled layers, it already includes any downstream gradient modifications. Using the same fixed basis as the diagnostic and a detached control flag, Alpha-Stabler passes upstream

$$\tilde{G}_\ell^{(t)} = G_\ell^{(t)} - a_\ell^{(t)} (G_\ell^{(t)} \mathcal{U}_\ell) \mathcal{U}_\ell^\top. \quad (67)$$

The actor’s forward computation is unchanged. At an active layer, Equation (67) is the orthogonal projection $G_\ell^{(t)} Q_\ell$; at an inactive layer, the incoming gradient is unchanged.

Proposition D.4 (Minimal-change removal of principal gradients). *For an orthogonal projector P and $Q = I - P$, the unique solution of*

$$\underset{Z: ZP=0}{\text{minimize}} \quad \frac{1}{2} \|Z - G_\ell\|_F^2 \quad (68)$$

is $\tilde{G}_\ell = G_\ell Q$. It satisfies

$$\tilde{G}_\ell P = 0, \quad \tilde{G}_\ell Q = G_\ell Q, \quad \|\tilde{G}_\ell\|_F^2 = \|G_\ell\|_F^2 - \|G_\ell P\|_F^2. \quad (69)$$

Proof. Every feasible Z obeys $Z = ZQ$. Since the principal and complement components are Frobenius-orthogonal, $\|Z - G_\ell\|_F^2 = \|Z - G_\ell Q\|_F^2 + \|G_\ell P\|_F^2$. The first term is uniquely minimized at $Z = G_\ell Q$. The remaining identities follow from $P^2 = P$, $Q^2 = Q$, and $PQ = 0$. $\square$

Active control therefore makes the smallest Euclidean change that removes the incoming gradient’s principal component, preserves its local complement, and cannot increase its norm at that site. The resulting signal is propagated through the network, and actor parameters are updated with the baseline optimizer.

From projected gradients to activation changes. We next relate this local gradient projection to the subsequent activation change through the parameter-to-activation Jacobian. The calculation considers one post-warm-up SGD update with a fixed reference projector.

Lemma D.7 (Local principal leakage under projected SGD). *Fix a token batch, one active control site, and a parameter block ϑ . Suppose the loss depends on this block only through the selected activations, other parameters are held fixed, and $h(\vartheta)$ is twice continuously differentiable locally. Stack token activations as $h = \text{vec}(H^\top)$, and define $J = \partial h / \partial \vartheta$, $\Pi = I_N \otimes P$, $\mathcal{Q} = I - \Pi$, and $g = \nabla_h \mathcal{L}$. For a plain SGD step with size α small enough that the update segment remains in a neighborhood with bounded second derivatives, using the projected gradient gives*

$$\delta \vartheta = -\alpha J^\top \mathcal{Q}g, \quad \delta h = -\alpha K \mathcal{Q}g + R, \quad K = J J^\top, \quad (70)$$

where $\|R\|_2 \leq c_h \alpha^2 \|J^\top \mathcal{Q}g\|_2^2$ for a local constant c_h . Therefore,

$$\|\Pi \delta h\|_2 \leq \alpha \|\Pi K \mathcal{Q}\|_2 \|\mathcal{Q}g\|_2 + c_h \alpha^2 \|J^\top \mathcal{Q}g\|_2^2. \quad (71)$$

Proof. The assumed dependency gives the modified parameter gradient $J^\top \mathcal{Q}g$. Taylor expansion yields $h(\vartheta + \delta \vartheta) - h(\vartheta) = J \delta \vartheta + R$, with the stated remainder bound from local second-derivative control. Substituting the SGD step and applying Π proves Equation (71) by submultiplicativity. $\square$

The first-order principal displacement is governed by the coupling $\Pi K \mathcal{Q}$. If this coupling vanishes, projection eliminates the first-order principal displacement from the selected parameter block. More generally, Equation (71) quantifies how the local projected gradient translates into principal activation displacement.

D.8 SUMMARY OF THEORETICAL AND EMPIRICAL RESULTS

Table 5 summarizes the theoretical and empirical results. Parts I–II characterize fixed-vector recovery and independently effective steering directions. Part III establishes the energy interpretation of PSI, the minimal-change property of gradient projection, and a local bound relating projected SGD to activation changes.

Summary. The analysis connects RLVR steering and geometry-guided control in three stages. Shared teacher-induced shifts explain when a fixed vector can recover substantial target energy; constrained fitting characterizes how additional interventions depend on the available directions and context dependence; and the Alpha-Stabler analysis gives an energy interpretation of PSI and exact local properties of backward projection. A fixed warm-up reference links the diagnostic to the control subspace, while the local coupling bound describes how projected gradients affect subsequent activations. Together, these results provide a mathematical framework for the empirical capacity, localization, and stabilization findings.

Table 5: Summary of recovery, steering geometry, and gradient-control results.

Statement	Reference	Scope / status
<i>Part I — fixed-vector recovery</i>		
Distillation admits a weighted least-squares approximation	Proposition D.1	Local KL and network approximations
KL-regularized optimum is an exponential reward tilt	Lemma D.2	Exact for the stated optimum, $\beta > 0$
Next-token probabilities follow continuation normalizers	Lemma D.3	Exact under the tilt model
Shared profiles and consistent pullbacks yield shared shifts	Lemmas D.4, D.5	Sufficient conditions
$\rho = \text{SNR}/(1 + \text{SNR})$	Theorem D.1	Constant-metric quadratic model
ρ is bounded by the leading spectral energy share	Corollary D.2	Upper bound; not general equality
$\rho \geq \kappa/[(1 + \eta)(1 + \varepsilon)]$	Theorem D.2	Stated geometric assumptions
<i>Part II — geometry of steering interventions</i>		
Constrained constant-vector recovery is a metric projection	Theorem D.3	Fixed-metric quadratic problem
Adaptive subspace recovery equals a spectral partial sum	Equation (61)	Context-dependent coefficients
Orthogonal directions retain standalone effectiveness	Section D.5	Empirical intervention comparison
Effective directions have low principal energy	Proposition D.3 and steering analysis	Metric-conditional theory; Euclidean measurements are distinct
Weight-space structure can bound activation leakage	Lemma D.6	Requires its explicit leakage condition
<i>Part III — monitoring and gradient control</i>		
PSI measures token-level Euclidean principal energy	Corollary D.4	Exact energy identity with numerical stabilization quantified
Gradient projection minimally removes principal components	Proposition D.4	Exact at each active hook with the projector fixed
Principal activation displacement is bounded by coupling	Lemma D.7	Single-site local SGD assumptions
Warm-up defines the reference basis and diagnostic thresholds	Algorithm 1	Estimated during warm-up and fixed thereafter
PSI provides useful warning and control improves stability	Figures 7(c) and 8	Empirical training evaluation